

University of St. Gallen

School of Management, Economics, Law, Social Sciences, International Affairs and
Computer Science

**Federal Reserve Communication and Financial Market
Reactions:**

Evidence from Agent-Specific Sentiment in FOMC Documents

Lionel Zaugg

lionel.zaugg@student.unisg.ch

20-938-320

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Prof. Ulrich Matter

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Abstract

This thesis examines whether agent-specific sentiment in Federal Reserve communication is associated with financial-market returns after FOMC communication events. Using CentralBankRoBERTa, document-level net sentiment scores are constructed for five economic-agent categories across 107 FOMC minutes, 86 press conference transcripts, and 110 policy statements from 2012 to 2025. Policy statements are used for descriptive analysis, while the regression analysis focuses on minutes and press conferences. An event-level local projection framework estimates the association between agent-specific sentiment and cumulative log returns of the S&P 500, two-year Treasury futures, and Bitcoin over horizons of zero to nine calendar days, with monetary-policy surprise controls included for press conference specifications. Descriptively, the sentiment series vary plausibly across communication formats and macroeconomic episodes, supporting their usefulness for organizing Federal Reserve communication. Empirically, the results are limited. Hypotheses 1 and 2, which predict that positive real-economy sentiment raises equity returns and lowers Treasury futures returns, are not supported; coefficients are weak, unstable, or point in the wrong direction. Hypothesis 3 receives partial support: financial-sector sentiment in press conferences is positively associated with Bitcoin returns at intermediate and longer horizons, a pattern that persists after controlling for monetary-policy surprises but does not extend to the S&P 500 or Treasury futures. Overall, the findings suggest that broad agent-specific sentiment categories are descriptively informative but not robustly priced across asset classes, and that future work should align classification objectives more tightly with specific transmission channels.

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List of Abbreviations

BERT	Bidirectional Encoder Representations from Transformers
BTC	Bitcoin
BTC/USD	Bitcoin/U.S. dollar exchange rate
CentralBankRoBERTa	RoBERTa-based language model fine-tuned on central bank texts
ECB	European Central Bank
Fed	Federal Reserve
FOMC	Federal Open Market Committee
GPT	Generative Pre-trained Transformer
LLM	Large language model
MPS	Monetary-policy surprise
QE	Quantitative easing
RoBERTa	Robustly Optimized BERT Pretraining Approach
S&P 500	Standard & Poor's 500 Index
SPX	S&P 500 index ticker abbreviation
ZT	Two-year Treasury futures ticker abbreviation

1 Introduction

Central bank communication has become a central instrument of modern monetary policy. Financial markets react not only to policy-rate decisions, but also to how central banks describe economic conditions, risks, and the likely future path of policy. At the same time, recent advances in machine learning have made it possible to extract structured information from economic texts at a level of detail that traditional dictionary-based methods cannot easily provide. This thesis combines these two developments by studying whether agent-specific sentiment in Federal Reserve communication contains economically relevant information for financial markets.

The empirical focus is on the Federal Reserve because of its central role in global financial markets and the international transmission of monetary policy (Chen et al., 2016; Dées & Galesi, 2021; Ehrmann & Fratzscher, 2009; Gilchrist et al., 2019; Kalemli-Özcan, 2019; Kazi et al., 2013; Miranda-Agrippino & Rey, 2020). Around Federal Open Market Committee meetings, the Fed communicates through several key formats. Policy statements are released immediately after each meeting and summarize the policy decision, forward guidance, and the Committee’s assessment of risks. FOMC minutes are published about three weeks later and provide a more detailed account of the discussion within the Committee. Press conferences, held after FOMC announcements, allow the Chair to explain the decision and respond to questions. These formats are central to how professional observers interpret Federal Reserve policy (Blinder et al., 2008). Recent evidence suggests that press conferences have become especially important for market participants (Acosta et al., 2025; Boguth et al., 2019; E. Swanson & Jayawickrema, 2023; E. T. Swanson, 2023).

This thesis analyzes FOMC policy statements, FOMC minutes, and press conference transcripts from 2012 to 2025. To measure the tone of communication, I use CentralBankRoBERTa (Pfeifer & Marohl, 2023), a transformer-based language model that classifies each sentence by economic-agent category (households, firms, government, the financial sector, and the central bank) and sentiment (positive or negative). Sentence-level classifications are aggregated into document-level net sentiment measures for each agent, as described in Chapter 3.2.

This thesis first assesses whether CentralBankRoBERTa produces economically interpretable sentiment measures when applied to Federal Reserve communication. Since the model assigns each sentence to one agent and one sentiment category, the descriptive analysis (Chapter 5) examines whether the resulting document-level measures behave plausibly across communication formats, agent categories, and major macroeconomic episodes.

Second, the thesis examines whether these sentiment measures are associated with financial-market reactions after FOMC minutes and press conferences. The empirical analysis links document-level agent-specific sentiment to cumulative event-window returns of the S&P 500, two-year Treasury futures, and Bitcoin, which represent equity-market exposure, short-term interest-rate-sensitive fixed-income exposure, and a digital asset with an ambiguous connection to conventional monetary-policy transmission. Policy statements are excluded from the regression analysis, as explained in Chapter 4.1.

The research question is: *How is agent-specific sentiment in Federal Reserve communication associated with financial-market returns after FOMC minutes and press conferences?* The question is formulated in terms of association rather than causality, because Federal Reserve sentiment is not randomly assigned and may reflect information already available to market participants. The empirical analysis uses event-

level local projections over several post-event horizons and includes monetary-policy surprise controls for press conferences where available.

The thesis contributes by applying an agent-specific sentiment framework to Federal Reserve communication, evaluating whether CentralBankRoBERTa produces economically meaningful measures in this setting, and testing whether these measures are associated with financial-market returns across equities, Treasury futures, and Bitcoin.

The remainder of the thesis is structured as follows. Chapter 2 reviews the literature on Federal Reserve communication, financial-market transmission, Bitcoin’s exposure to monetary-policy conditions, and textual analysis in economics and finance, before developing the conceptual framework and hypotheses. Chapter 3 describes the construction of the communication corpus, the sentiment variables, the financial-market returns, and the final event-level dataset. Chapter 4 presents the empirical strategy used to test the hypotheses. The subsequent chapters report the descriptive text-analysis results (Chapter 5), the local-projection estimates (Chapter 6), the robustness checks (Chapter 7), and the conclusion (Chapter 8).

2 Literature Review

2.1 Federal Reserve Communication as a Monetary Policy Tool

“As I had often remarked, monetary policy is 98 percent talk and 2 percent action.” (Bernanke, 2015, p. 498)

The Federal Reserve’s communication strategy has shifted over the past four decades from opacity to active expectation management (Bernanke, 2020; Blinder et al., 2008; Woodford, 2005).

Before the 1990s, the Fed provided almost no explicit information, and markets inferred policy from operations (Woodford, 2005). This changed in 1994, when the FOMC began announcing its federal funds rate target after meetings, marking the first systematic disclosure of policy decisions (Woodford, 2005).

In the late 1990s, the Fed introduced qualitative forward-looking signals (Woodford, 2005). A “policy bias” in 1998 indicated the likely direction of the next rate move, while the “balance of risks” in 1999 shifted communication toward a broader assessment of inflation versus growth risks, guiding expectations more indirectly (Woodford, 2005).

By the early 2000s, communication evolved into explicit forward guidance. Statements such as maintaining accommodation for a “considerable period” signaled that rates would remain low longer than usual, conditional on economic conditions, thereby influencing the expected path of policy. At the same time, transparency increased through faster publication of FOMC minutes (Blinder et al., 2008; Woodford, 2005).

The shift toward greater transparency was driven by a parallel change in the theoretical understanding of how monetary policy works. Early contributions highlighted the credibility problem inherent in discretionary policy. Kydland and Prescott (1977) showed that without commitment, policymakers face a time-inconsistency problem that leads to systematically higher inflation. Barro and Gordon (1983) extended this insight by emphasizing the role of reputation and expectations. These frameworks implied that policy effectiveness depends critically on how private agents form beliefs about future actions, creating a rationale for clearer communication to anchor expectations.

A second strand of the literature introduced imperfect information and learning. Cukierman and Meltzer (1986) modeled central banks as possessing private information, implying that communication becomes a strategic choice. Later work on adaptive learning, such as Orphanides and Williams (2004), showed that when agents must infer the central bank’s objectives from observed behavior, poor communication can destabilize expectations. In this setting, transparency improves macroeconomic outcomes by helping the public correctly understand the policy framework (Blinder et al., 2008).

The most decisive shift came with the New Keynesian framework, particularly the work of Michael Woodford. Woodford and Walsh (2005) and Woodford (2005) formalized the idea that monetary policy operates primarily through expectations of future interest rates rather than current rates. Because long-term rates depend on expected future short rates, communication about future policy becomes a central transmission channel. In this view, transparency is not merely about accountability but is essential for policy effectiveness itself (Woodford, 2005).

A complementary insight comes from coordination models. Morris and Shin (2002) showed that public signals can coordinate private expectations, giving central bank communication a disproportionate influence on market outcomes. While this raised concerns about potential overreliance on central bank

signals, it reinforced the idea that communication is a powerful policy tool.

The global financial crisis marked a decisive turning point that elevated communication from a supporting tool to a central instrument of policy. At the effective lower bound, conventional rate policy became constrained, and models such as Eggertsson and Woodford (2003) showed that credible commitments about future policy—forward guidance—could substitute for current rate cuts by shaping expectations of the future path of interest rates. This mechanism was actively used by the Fed after 2008, making communication itself a primary channel of monetary transmission (Bernanke, 2020).

At the same time, the crisis exposed limitations of the pre-crisis framework, which had focused narrowly on inflation stabilization while largely neglecting financial imbalances. Contributions such as Blanchard (2012) and Borio (2014a, 2014b) argued that monetary policy must incorporate financial stability considerations, while Woodford (2012) revisited inflation-targeting frameworks in light of these challenges. Empirical and historical work, such as Taylor (2015) and Mian and Sufi (2015), further emphasized the macroeconomic importance of credit cycles and balance-sheet dynamics.

These developments had direct implications for communication. Central banks increasingly used communication not only to guide interest-rate expectations but also to explain risk assessments, unconventional policies, and interactions with financial-stability tools (Blanchard, 2012; Borio, 2014a, 2014b). In particular, the rise of forward guidance and balance-sheet policies required central banks to communicate explicitly about the future policy path and the conditions under which policies would change, thereby making communication more forward-looking and state-contingent (Bernanke, 2020; Woodford, 2012). At the same time, the growing emphasis on financial cycles and macroprudential policy broadened the scope of communication beyond inflation and output stabilization (Mian & Sufi, 2015; Taylor, 2015). This shift is reflected in later assessments, such as Borio (2024), which characterize modern monetary policy as a combination of expectations management, balance-sheet policies, and macro-financial surveillance.

In short, while earlier theory established why transparency improves policy effectiveness (Blinder et al., 2008; Woodford, 2005), the post-crisis period demonstrated that communication can itself function as a primary policy instrument. This is particularly evident at the effective lower bound, where shaping expectations about future policy became a central channel of monetary transmission (Bernanke, 2020; Eggertsson & Woodford, 2003), and in environments with elevated financial risks, where communication helps integrate monetary and financial stability objectives (Borio, 2024; Borio, 2014a, 2014b).

2.2 Transmission of Central Bank Communication to Financial Markets

Monetary policy is transmitted to financial markets mainly through several distinct but overlapping channels. The first is the surprise or target-rate channel: asset prices react primarily to the unexpected component of policy decisions, not to anticipated changes. Kuttner (2001) established this using federal funds futures.

A second channel is the expectations or forward-guidance channel. Because long-term rates depend on expected future short rates, communication about the future policy path can move yields even without an immediate rate change (Gurkaynak et al., 2005; Woodford & Walsh, 2005). Gurkaynak et al. (2005) showed that FOMC announcements contain both a current-rate “target” factor and a forward-looking “path” factor, with the latter especially important for longer-term yields and asset prices. This channel became central at the effective lower bound, where forward guidance and asset purchases substituted for

conventional rate cuts (Bernanke, 2020; Eggertsson & Woodford, 2003). Bernanke (2020) argues that forward guidance works partly by committing policymakers to “lower-for-longer” policy, while QE works through both signaling and portfolio-balance effects.

A third channel is the risk-premium channel: asset prices reflect discounted expected cash flows, where the discount rate consists of both expected future short rates and risk premia (Campbell & Shiller, 1988; Cochrane, 2011). In bond markets, yields can be decomposed into expected short rates and term premia (Adrian et al., 2013), and monetary policy affects long-term yields in part through changes in the term premium (Hanson & Stein, 2015). Empirically, Bernanke and Kuttner (2005) show that stock market reactions to monetary policy surprises are largely driven by changes in expected excess returns, indicating that policy shifts affect equity risk premia rather than only the risk-free rate. Similarly, Gilchrist et al. (2015) show that monetary policy shocks significantly affect corporate credit spreads, linking policy announcements to firms’ external financing conditions. In fixed-income markets, unconventional policy such as large-scale asset purchases compresses term premia through portfolio-balance effects (Bernanke, 2020). Together, these results imply that monetary policy influences asset prices not only via expected future short rates but also through changes in risk premia across asset classes.

Fourth, announcements may reveal not only policy intentions but also information about the economic outlook (Blinder et al., 2008; Jarociński & Karadi, 2020; Nakamura & Steinsson, 2018; Romer & Romer, 2000). Romer and Romer (2000) show that Federal Reserve forecasts contain information beyond private forecasts, while Nakamura and Steinsson (2018) argue that FOMC announcements can shift beliefs about economic fundamentals, not only about policy. Jarociński and Karadi (2020) further distinguish between pure monetary policy shocks and information shocks using the joint response of interest rates and stock prices. Bauer and Swanson (2023), however, offer an alternative interpretation, arguing that the observed comovement may reflect the Fed reacting more strongly to publicly available information than markets had anticipated, rather than revealing private information.

A fifth channel is the textual communication channel. Empirical evidence shows that variation in the wording and framing of central bank communication can affect financial markets even after controlling for policy decisions and observable information. Hansen and McMahon (2016) identify “language shocks” in central bank communication and show that these have independent effects on macroeconomic and financial variables. Similarly, Cieslak and Schrimpf (2019) decompose central bank communication into monetary and non-monetary components and show that these components have distinct effects on asset prices, indicating that communication contains variation beyond pure policy signals. Tobback et al. (2017) and Hubert and Labondance (2018) further demonstrate that linguistic features such as hawkish/dovish framing and sentiment systematically influence policy expectations and financial variables. Together, these findings imply that textual variation—conditional on policy actions and information—constitutes a distinct transmission channel, typically identified using text-as-data methods (Gentzkow et al., 2019).

A sixth channel is the vocal-tone channel. Conditional on policy decisions and the informational content of communication, the non-verbal delivery of central bank communication can independently affect financial markets. Gorodnichenko et al. (2023) show that the emotional tone of Federal Reserve Chairs’ speech—measured from acoustic features such as pitch and frequency—has significant effects on asset prices, even after controlling for monetary policy shocks and textual sentiment. In particular, a more positive tone raises equity prices and reduces perceived risk, indicating that markets extract “soft information” from vocal delivery. Consistent evidence is provided by Cho (2026), who finds that vocal

tone in monetary-policy communication affects cryptocurrency prices. Taken together, this literature identifies a distinct transmission channel in which markets respond not only to what central banks say or do, but also to how policy is communicated.

Overall, the literature suggests that financial-market transmission runs through expected future short rates, risk premia, credit conditions, portfolio rebalancing, information revelation, and—beyond the substantive content of communication—the textual framing and vocal delivery of central bank announcements. The main unresolved issue is identification: observed asset-price movements around announcements may combine a pure policy shock with information about the central bank’s economic outlook.

2.3 The Transmission of Monetary Policy for the Case of Bitcoin

The transmission channels discussed in the previous section are most direct for Treasury securities and equities, but recent literature suggests that they may also be relevant for Bitcoin. Although Bitcoin is not issued by a central bank and does not generate contractual cash flows, its price may still respond to monetary policy through liquidity conditions, risk appetite, opportunity costs, and expectations about future demand. This possibility is especially relevant because Bitcoin has increasingly been studied not only as a payment technology, but also as a speculative and financial asset (Baur et al., 2018; Biais et al., 2023; Liu et al., 2022).

Early empirical work suggests that Bitcoin initially differed strongly from conventional monetary and financial assets. Baur et al. (2018) show that Bitcoin was used primarily as a speculative investment rather than as a medium of exchange and that its returns were largely uncorrelated with traditional asset classes. This implies that, in its earlier phase, Bitcoin was relatively detached from the standard monetary-policy transmission channels that operate through interest rates, expected cash flows, and portfolio rebalancing. At the same time, the speculative character of Bitcoin makes it potentially sensitive to changes in liquidity and risk appetite, since its valuation depends less on stable income streams than on expectations about future demand and adoption (Baur et al., 2018).

Theoretical and empirical work increasingly supports the view that Bitcoin can be understood as part of a broader financial asset universe. Biais et al. (2023) develop an equilibrium model in which Bitcoin has a fundamental component linked to transactional usefulness, but where prices are also strongly affected by self-fulfilling beliefs and extrinsic volatility. This suggests that Bitcoin is neither a pure currency nor a pure bubble, but an asset whose price can react strongly to changes in beliefs and market conditions. Liu et al. (2022) show that cryptocurrency returns exhibit a systematic factor structure, with market, size, and momentum factors helping to explain expected returns. This finding is important for the present thesis because it suggests that cryptocurrencies are not purely idiosyncratic assets, but increasingly display systematic risk components that can be analyzed with asset-pricing tools (Liu et al., 2022).

Monetary policy may affect Bitcoin through several channels. Contractionary monetary policy can raise the opportunity cost of holding non-yielding assets, tighten financial conditions, and reduce demand for speculative assets. Expansionary monetary policy can work in the opposite direction by supporting liquidity and risk-taking. These mechanisms are not identical to the transmission of monetary policy to bonds or equities, because Bitcoin has no coupon, dividend, or direct claim on future cash flows. However, they are consistent with the view that Bitcoin’s valuation is sensitive to changes in investor beliefs, market liquidity, and risk appetite (Baur et al., 2018; Biais et al., 2023).

Empirical evidence increasingly points toward monetary-policy sensitivity in Bitcoin markets, although the findings are not uniform. Ma et al. (2022) show that unexpected U.S. monetary tightening lowers Bitcoin prices around FOMC announcements, suggesting that Bitcoin reacts to monetary policy surprises in a way that is consistent with risky-asset behavior. Karau (2023) further shows that Bitcoin’s response to U.S. monetary policy has changed over time. In earlier periods, monetary tightening could increase Bitcoin demand through alternative channels, while in the post-2020 period contractionary monetary policy is more clearly associated with lower Bitcoin prices. This supports the interpretation that Bitcoin’s sensitivity to U.S. monetary policy has increased as the market has matured and become more integrated into broader financial markets (Karau, 2023; Ma et al., 2022).

However, the literature does not imply that Bitcoin reacts to macroeconomic and monetary news in the same way as traditional assets. Benigno and Rosa (2023) find that Bitcoin is largely disconnected from many U.S. macroeconomic and monetary news announcements in intraday event windows, with the exception of CPI news. This provides an important counterpoint to studies that find stronger monetary-policy effects. It suggests that Bitcoin’s macro-financial sensitivity depends on the sample period, event horizon, type of announcement, and identification strategy. Therefore, the transmission of monetary policy to Bitcoin should be understood as conditional rather than mechanical (Benigno & Rosa, 2023; Karau, 2023; Ma et al., 2022).

For this thesis, the relevant implication is that Bitcoin provides a useful test case for whether Federal Reserve agent-specific communication affects assets beyond traditional equity and bond markets. If Bitcoin reacts to sentiment in Federal Reserve communication, this would suggest that the market prices not only formal policy surprises but also broader communication signals about financial conditions, risk appetite, and the policy outlook. A response similar to the S&P 500 would support the interpretation of Bitcoin as a risky asset exposed to macro-financial sentiment. A weak or unstable response would instead be consistent with the view that Bitcoin remains only partially integrated into conventional monetary-policy transmission (Benigno & Rosa, 2023; Karau, 2023; Liu et al., 2022; Ma et al., 2022).

2.4 Development of Textual Analysis in Economics and Finance

Early contributions in finance demonstrated that textual information contains economically relevant signals: Tetlock (2007) shows that media pessimism predicts stock market movements, while Tetlock et al. (2008) link firm-specific news content to earnings and returns. In accounting, Li (2008) establishes that the readability of annual reports is systematically related to firm performance. These studies rely on word counts and predefined lexicons, implicitly assuming that meaning can be approximated by the frequency of individual terms.

A second stage refined these approaches by introducing domain-specific dictionaries and more structured textual measures. Loughran and McDonald (2011) demonstrate that general-purpose sentiment dictionaries misclassify many financial terms, leading to systematic bias, and propose a finance-specific lexicon. In parallel, text-based measures expanded into macroeconomics and policy analysis. Baker et al. (2016) construct an index of economic policy uncertainty from newspaper coverage, while Cieslak and Schrimpf (2019) show that central bank communication contains both monetary and non-monetary components with distinct financial-market effects. This literature establishes that textual data can be used to measure expectations, uncertainty, and policy signals, but it remains largely based on bag-of-words

representations that ignore context and sentence structure (Gentzkow et al., 2019).

The limitations of bag-of-words approaches—particularly their inability to capture context, semantics, and word interactions—motivated a shift toward machine-learning-based text analysis (Gentzkow et al., 2019). Early machine-learning applications, including topic models (Blei et al., 2003) and supervised classifiers, improved predictive performance but still relied on sparse representations that struggle with contextual meaning (Gentzkow et al., 2019). In central bank communication, this limitation is especially binding, as policy statements rely heavily on conditional, forward-looking, and context-dependent language.

A major methodological breakthrough came with the introduction of transformer-based language models. Devlin et al. (2019) introduce BERT, which uses bidirectional attention to generate contextual embeddings, and Liu et al. (2019) show that improved pre-training strategies, as in RoBERTa, further enhance performance. These models outperform previous approaches across a wide range of natural language processing tasks because they capture context and long-range dependencies in text (Devlin et al., 2019; Liu et al., 2019).

In finance, this paradigm shift led to the development of domain-specific transformer models. Araci (2019) introduces FinBERT, showing that a model pre-trained on general language and fine-tuned on financial text significantly outperforms traditional lexicon-based sentiment methods. Subsequent empirical work indicates a shift toward transformer-based models in financial text analysis, which consistently outperform earlier approaches in domain-specific tasks (Araci, 2019; Mishev et al., 2020; Pfeifer & Marohl, 2023).

These advances are particularly relevant for central bank communication. Monetary policy texts are characterized by technical language, forward guidance, and subtle shifts in tone, which require context-sensitive interpretation (Hansen & McMahon, 2016). Empirical evidence shows that textual variation in central bank communication has measurable macroeconomic and financial effects, as discussed in Chapter 2.2. Recent work using transformer-based methods further shows that contextual models outperform lexicon-based approaches in analyzing policy communication and allow for more granular representations of sentiment, topics, and economic entities (Araci, 2019; Pfeifer & Marohl, 2023).

The most recent stage in this development is the adaptation of transformer models specifically to central bank communication. Building on architectures such as RoBERTa (Liu et al., 2019) and domain-specific models such as FinBERT (Araci, 2019), recent work fine-tunes language models on central bank corpora. CentralBankRoBERTa represents this approach: it is based on a RoBERTa architecture and further fine-tuned on central bank texts to capture policy-specific language and communication patterns (Pfeifer & Marohl, 2023). Domain adaptation of this type has been shown to improve performance in specialized text domains by aligning model representations with domain-specific terminology and structure (Gururangan et al., 2020; Pfeifer & Marohl, 2023). This enables the model to distinguish subtle variations in policy tone and interpret complex, conditional statements characteristic of monetary-policy communication.

Overall, the evolution of textual analysis in economics and finance can be understood as a transition from word-count-based proxies to context-aware representation learning. For central bank communication in particular, this progression culminates in domain-specific transformer models such as CentralBankRoBERTa, which combine general language understanding with specialized knowledge of monetary policy communication.

2.5 Conceptual Framework and Hypotheses

The literature reviewed above suggests that Federal Reserve communication should be understood as an information event rather than as a simple description of current policy decisions. FOMC communication can shape expectations about the future policy path, reveal the central bank’s assessment of economic and financial conditions, affect risk premia, and coordinate market beliefs around a common public signal (Blinder et al., 2008; Gurkaynak et al., 2005; Hansen & McMahon, 2016; Jarociński & Karadi, 2020; Nakamura & Steinsson, 2018; Woodford, 2005). The empirical question of this thesis is therefore whether agent-specific sentiment extracted from Federal Reserve communication contains economically relevant information and whether this information is associated with financial-market returns after communication events.

The conceptual framework has two parts. First, the thesis assesses whether CentralBankRoBERTa produces interpretable measures of Federal Reserve communication. This descriptive validation is important because central bank language is often ambiguous: a single sentence may be relevant for several agents at once, and positive sentiment may have different implications depending on the economic context. If the document-level sentiment measures capture economically meaningful information, they should display plausible variation across communication formats, agent categories, and major macroeconomic episodes.

Second, the thesis examines whether these sentiment measures are associated with market reactions. Positive sentiment about households and firms can be interpreted as favorable real-economy sentiment. Such communication may signal stronger consumption, employment, investment, or business conditions. For equities, the unexpected component of favorable real-economy sentiment can be positive through higher expected cash flows and lower macroeconomic risk premia (Bernanke & Kuttner, 2005; Campbell & Shiller, 1988; Cochrane, 2011). Because financial markets are forward-looking and prices reflect all publicly anticipated information, only the surprise component of FOMC communication should generate a systematic return response (Kuttner, 2001). At the same time, stronger real-economy sentiment may also imply a less accommodative future policy path, which can raise discount rates. The expected equity-market effect is therefore not mechanically positive, but the baseline hypothesis is that favorable real-economy sentiment is associated with higher S&P 500 returns if markets interpret its unexpected component primarily as good news about growth and risk.

H1: More positive household and firm sentiment in Federal Reserve communication is associated with higher S&P 500 returns after communication events.

For short-term interest-rate-sensitive assets, the expected mechanism differs. If the unexpected component of positive household or firm sentiment leads investors to revise upward their expectations of future short-term rates, two-year Treasury yields should rise. Since this thesis uses returns on two-year Treasury futures rather than yields, the expected sign is negative because Treasury futures prices generally move inversely to yields. This motivates the second hypothesis.

H2: More positive household and firm sentiment in Federal Reserve communication is associated with lower two-year Treasury futures returns after communication events.

Financial-sector sentiment may be relevant through a different channel. The unexpected component of sentiment about the financial sector captures markets’ surprise at the Federal Reserve’s assessment of

financial conditions, market functioning, stress, and risk appetite. This category is conceptually close to the pricing of risk-bearing assets. The S&P 500 is exposed to earnings expectations, discount rates, and equity risk premia. Bitcoin, whose sensitivity to monetary policy and macro-financial conditions is reviewed in Chapter 2.3, provides a useful test case for whether Federal Reserve communication is relevant beyond traditional asset classes. The baseline hypothesis is that favorable financial-sector sentiment is associated with positive returns for both the S&P 500 and Bitcoin, as improved financial conditions reduce risk premia and support risk-on assets.

H3: Financial-sector sentiment in Federal Reserve communication is more strongly associated with risky-asset returns, especially the S&P 500 and Bitcoin, than with two-year Treasury futures returns.

The empirical analysis distinguishes textual sentiment from conventional interest-rate surprises—measured from federal funds futures—where possible, as described in Chapter 3.4.

The hypotheses are interpreted as conditional associations rather than causal effects. Federal Reserve sentiment is not randomly assigned and may partly reflect macroeconomic or financial information already known to market participants. If sentiment is nevertheless systematically associated with post-event returns, this may indicate that Federal Reserve communication contains additional information, changes the framing of existing information, or coordinates market beliefs around a common public signal. Conversely, weak or unstable coefficients would suggest that the extracted sentiment measures are either already priced, too noisy, or not central to market reactions at the horizons studied in this thesis.

3 Data

This chapter describes the construction of the event-level dataset used in the descriptive and empirical analysis. The dataset combines document-level agent-specific sentiment measures, financial-market returns, and, where available, monetary-policy surprise controls. Each observation corresponds to one Federal Reserve communication event.

3.1 Federal Reserve Communication Data

The textual corpus used for this thesis consists of FOMC policy statements, FOMC meeting minutes, and FOMC press conference transcripts from 2012 to 2025. All documents were collected from the Federal Reserve’s public website. The document-level language-model outputs contain 110 policy statements, 107 FOMC minutes, and 86 press conference transcripts. Three policy statements were unscheduled and therefore have no corresponding minutes document. Five scheduled policy statements and minutes documents from the sample period could not be included because they were not successfully retrieved during the automated scraping process and were not added manually. Since this exclusion reflects data-collection limitations rather than document content, it is not expected to create systematic selection based on communication tone.

3.2 Text Processing and Sentiment Construction

All raw texts were cleaned before applying CentralBankRoBERTa. For policy statements and minutes, the cleaning routines remove non-substantive material such as document headings, meeting labels, release-time stamps, voting records, footnotes, administrative sections, and duplicated policy-statement text embedded in the minutes. The purpose is to retain the substantive communication content relevant for sentiment classification.

Press conference transcripts require a separate procedure because they originate from PDF transcripts and contain both Chair and reporter turns. The cleaning pipeline corrects encoding artifacts, removes repeated page headers and page counters, trims each transcript to the first Chair opening line, reflows PDF line breaks, and normalizes whitespace. Reporter questions are removed using speaker labels, while Chair or Chairman turns are retained. The resulting press conference sentiment variables therefore summarize the Chair’s opening statement and answers, not the wording of journalist questions.

The main explanatory variables are document-level agent-specific net sentiment measures. For each document and agent category, net sentiment is calculated as the share of positive sentences minus the share of negative sentences among all sentences assigned to that agent:

$$s_{a,d} = \frac{N_{a,d}^{\text{pos}}}{N_{a,d}} - \frac{N_{a,d}^{\text{neg}}}{N_{a,d}},$$

where $s_{a,d}$ denotes net sentiment for agent a in document d , $N_{a,d}^{\text{pos}}$ and $N_{a,d}^{\text{neg}}$ denote the number of positive and negative sentences assigned to that agent, and $N_{a,d}$ denotes the total number of sentences assigned to that agent. The measure ranges from -1 to $+1$, with higher values indicating more positive tone. This procedure produces five document-level variables: $s_{\text{Households}}$, s_{Firms} , $s_{\text{FinancialSector}}$, $s_{\text{Government}}$, and $s_{\text{CentralBank}}$.

3.3 Financial-Market Data and Return Construction

The market data cover three assets: the S&P 500 (spx), two-year Treasury futures (zt), and Bitcoin (btc). S&P 500 and two-year Treasury futures prices are obtained from Yahoo Finance. Bitcoin prices are based on BTC/USD data from Bitstamp. Returns are constructed from daily open and close prices. The event-day baseline price is defined as the opening price on the communication date. The target price is the last available closing price on or before the relevant horizon date.

The dependent variables are cumulative event-window log returns. The return at horizon h is defined as

$$r_h = \log \left(\frac{P_{t+h}^{\text{close}}}{P_t^{\text{open}}} \right),$$

where P_t^{open} is the opening price on the event date and P_{t+h}^{close} is the last available closing price on or before the event date plus h calendar days. The horizons are $h \in \{0, 1, 2, 3, 5, 7, 9\}$. Thus, r_0 is the event-day open-to-close return, while longer horizons measure cumulative returns from the event-day open to the respective later close.

The horizons are defined in calendar days rather than trading days. If the target date falls on a weekend or market holiday, the most recent available trading close before that date is used.

3.4 Monetary-Policy Surprise Controls and Final Dataset

The thesis uses monetary-policy surprise controls for press conference regressions. The surprise series is obtained from the Federal Reserve Bank of San Francisco and is based on Bauer and Swanson (2023). The surprise variable captures the unexpected component of the corresponding FOMC policy announcement and covers the period from 2012 to 2023.

The final event-level datasets are retained separately for minutes and press conferences because the timing of communication and the availability of monetary-policy surprise controls differ across formats. Each observation contains the communication type, event date, document-level agent-specific sentiment variables, asset returns over the event windows, and, for press conferences where available, the monetary-policy surprise control.

4 Methodology

This chapter describes the empirical strategy used to examine whether agent-specific sentiment in Federal Reserve communication is associated with financial-market returns. Specifically, event-level local projection regressions relate document-level sentiment to post-event returns of the S&P 500, two-year Treasury futures, and Bitcoin.

The regression analysis focuses on FOMC minutes and press conference transcripts. Policy statements are included in the descriptive text analysis (Chapter 5) but excluded from the regression analysis because their short and standardized format makes agent-specific document-level sentiment comparatively noisy. All regressions are estimated separately by communication type and return horizon. This allows the association between Federal Reserve sentiment and asset returns to differ across minutes and press conferences, as well as across short and longer post-event windows.

4.1 Empirical Strategy

The empirical design follows an event-level local projection framework. Each observation corresponds to one Federal Reserve communication event indexed by i . The dependent variable is the cumulative log return of asset m over horizon h , where

$$h \in \{0, 1, 2, 3, 5, 7, 9\}.$$

The return horizons are defined in Chapter 3. The main explanatory variables are document-level net sentiment scores for the relevant economic-agent categories. These sentiment variables are denoted by s_i^k , where k refers to the economic-agent category.

For FOMC minutes, the baseline specification is estimated without monetary-policy surprise controls. Minutes are released several weeks after the corresponding FOMC decision and therefore do not correspond to the same announcement-window policy shock. For press conferences, the specification includes a monetary-policy surprise control where available, because press conferences take place shortly after the FOMC policy announcement on the same day. The press conference specification therefore separates the association between textual agent sentiment and asset returns from the conventional policy-surprise channel, although it does not eliminate all identification concerns.

The general minutes specification is

$$r_{i,h}^m = \alpha_h + \beta_h s_i^k + \varepsilon_{i,h},$$

where $r_{i,h}^m$ denotes the cumulative log return of asset m after communication event i over horizon h , s_i^k denotes document-level sentiment for agent category k , and $\varepsilon_{i,h}$ is the error term. The coefficient of interest is β_h , which measures the association between agent-specific sentiment and returns at horizon h .

For press conferences, the corresponding specification is

$$r_{i,h}^m = \alpha_h + \beta_h s_i^k + \gamma_h MPS_i + \varepsilon_{i,h},$$

where MPS_i denotes the monetary-policy surprise associated with the corresponding FOMC announcement. The coefficient γ_h captures the association between the conventional policy surprise and asset

returns, while β_h captures the conditional association between communication sentiment and returns.

As noted in Section 2.5, the estimates are interpreted as conditional associations rather than causal effects and should not be read as a fully identified causal effect of sentiment on asset prices.

4.2 Hypothesis 1: Real-Economy Sentiment and Equity Returns

H1 (Section 2.5) predicts that more positive household and firm sentiment is associated with higher S&P 500 returns, with the sign depending on whether the cash-flow channel or the discount-rate channel dominates. The specifications are estimated separately for minutes and press conferences.

For FOMC minutes, the specification is

$$r_{i,h}^{SPX} = \alpha_h + \beta_h s_i^k + \varepsilon_{i,h}, \quad k \in \{\text{Households, Firms}\}.$$

For press conferences, the corresponding specification is

$$r_{i,h}^{SPX} = \alpha_h + \beta_h s_i^k + \gamma_h MPS_i + \varepsilon_{i,h}, \quad k \in \{\text{Households, Firms}\}.$$

Here, $r_{i,h}^{SPX}$ denotes the S&P 500 return after communication event i over horizon h . The coefficient of interest is β_h . A positive estimate would be consistent with the interpretation that real-economy sentiment is priced mainly as favorable growth or risk-premium news. A weak or unstable estimate would indicate that the equity-market implications of real-economy sentiment are ambiguous or already incorporated into prices.

4.3 Hypothesis 2: Real-Economy Sentiment and Treasury Futures Returns

H2 (Section 2.5) predicts that more positive household and firm sentiment is associated with lower two-year Treasury futures returns, through the interest-rate channel. The specifications are estimated separately for minutes and press conferences.

For FOMC minutes, the specification is

$$r_{i,h}^{ZT} = \alpha_h + \beta_h s_i^k + \varepsilon_{i,h}, \quad k \in \{\text{Households, Firms}\}.$$

For press conferences, the corresponding specification is

$$r_{i,h}^{ZT} = \alpha_h + \beta_h s_i^k + \gamma_h MPS_i + \varepsilon_{i,h}, \quad k \in \{\text{Households, Firms}\}.$$

Here, $r_{i,h}^{ZT}$ denotes the two-year Treasury futures return after communication event i over horizon h . The coefficient of interest is again β_h . A negative estimate would be consistent with the interpretation that positive real-economy sentiment raises expected short-term interest rates. A positive, weak, or unstable estimate would suggest that the narrow interest-rate interpretation does not dominate the market response.

4.4 Hypothesis 3: Financial-Sector Sentiment Across Asset Classes

H3 (Section 2.5) is comparative rather than purely directional: it asks whether financial-sector sentiment is more strongly associated with risk-bearing assets, namely the S&P 500 and Bitcoin, than with two-year Treasury futures, given that the former are more sensitive to financial conditions and risk appetite. The specifications are estimated separately for minutes and press conferences.

For FOMC minutes, the specification is

$$r_{i,h}^m = \alpha_h + \beta_h s_i^{FinancialSector} + \varepsilon_{i,h}, \quad m \in \{SPX, BTC, ZT\}.$$

For press conferences, the corresponding specification is

$$r_{i,h}^m = \alpha_h + \beta_h s_i^{FinancialSector} + \gamma_h MPS_i + \varepsilon_{i,h}, \quad m \in \{SPX, BTC, ZT\}.$$

Here, $s_i^{FinancialSector}$ denotes document-level financial-sector sentiment. The hypothesis is evaluated by comparing the sign, magnitude, statistical significance, and stability of β_h across assets, horizons, and communication formats. A stronger and more stable association for the S&P 500 or Bitcoin than for two-year Treasury futures would be consistent with the interpretation that financial-sector sentiment is more relevant for risk-bearing assets than for short-term interest-rate futures. Differences between the S&P 500 and Bitcoin are also informative, because they indicate whether the sentiment measure captures broad risk appetite or a narrower form of macro-financial information that is more relevant for speculative assets.

4.5 Estimation and Interpretation

All regressions are estimated using ordinary least squares with heteroskedasticity-robust standard errors. The regressions are estimated separately by communication type, asset, sentiment category, and return horizon. This flexible setup avoids imposing a common response across minutes and press conferences or across assets with different transmission mechanisms.

The coefficient β_h is interpreted as the association between a one-unit increase in net sentiment and the cumulative return over horizon h . Since the sentiment variables range from -1 to $+1$, a one-unit increase represents a large change in the tone of communication. The economic interpretation of the coefficient should therefore take into account both the scale of the sentiment variable and the observed variation in the data.

For the S&P 500 and Bitcoin, a positive coefficient indicates that more positive sentiment is associated with higher asset prices over the relevant horizon. For two-year Treasury futures, the interpretation differs because futures prices generally move inversely to yields. A negative coefficient for two-year Treasury futures is consistent with a rise in expected short-term interest rates, while a positive coefficient is consistent with higher futures prices and lower implied yields.

The results are evaluated using statistical significance, economic magnitude, sign consistency across horizons, stability across communication formats, and consistency with the theoretical channels discussed in the literature review. Since the purpose of the analysis is not to identify a structural causal effect, isolated significant coefficients are interpreted cautiously. Greater weight is placed on patterns that are coherent across adjacent horizons and robust to the inclusion of monetary-policy surprise controls.

5 Descriptive Text-Analysis Results

The first descriptive finding is that document-level sentiment requires enough classified sentences to be interpretable. Since document net sentiment is aggregated from sentence-level classifications, short documents can produce volatile scores: one or two sentence-level classifications may strongly affect the document-level measure.

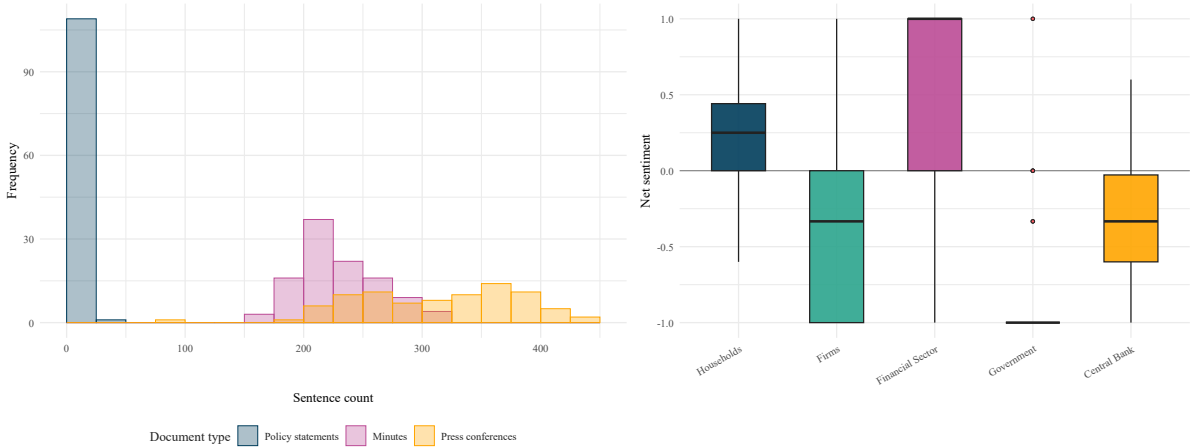


Figure 1: Sentence-count distribution and policy-statement net sentiment

This is most relevant for policy statements. Almost all policy statements contain fewer than 25 classified sentences, while minutes and press conferences contain substantially more text. The problem is visible in the sentiment distribution: policy statements show large variation across agents, with many values close to the bounds of the scale.

For policy statements, this variation is difficult to interpret because it may reflect the small number of classified sentences rather than meaningful differences in communication. For this reason, policy statements are excluded from further analysis and the following interpretation therefore focuses on minutes and press conferences.

Next, the distribution of classified sentences provides a first indication that the model captures meaningful structure in Federal Reserve communication. Minutes contain a relatively balanced distribution across households, firms, the financial sector, and the central bank. This is plausible because minutes summarize the Committee’s broader assessment of economic conditions, including consumption, labor markets, business activity, financial conditions, and policy considerations. Press conferences, by contrast, assign a larger share of sentences to the central bank. This also fits the role of press conferences. They are used to explain the policy decision, clarify the Committee’s assessment, and discuss the expected path of monetary policy. This is especially true for the Q&A section, where the Chair often explains or qualifies the Committee’s decision. Since the Chair speaks on behalf of the central bank, many of these sentences are assigned to the central-bank category.

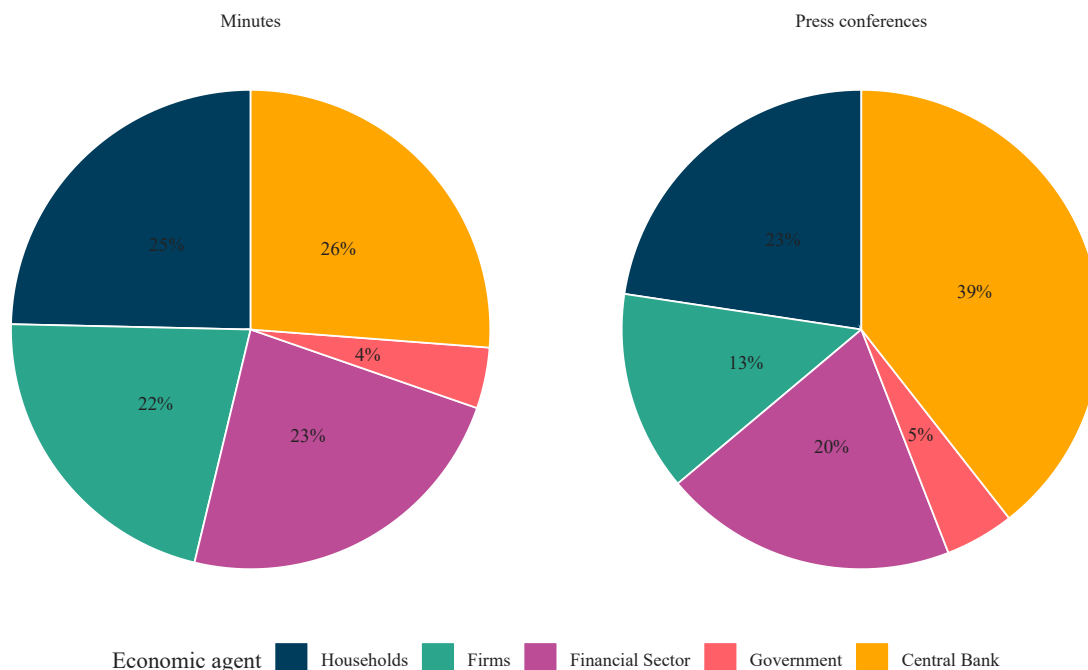


Figure 2: Agent classification shares in minutes and press conferences

To further assess the classification output, Figure 3 shows an excerpt from the press conference on 30 October 2019. The full sentence-level visualization of this press conference is provided in Appendix C. The example illustrates both the usefulness and the limits of the model. CentralBankRoBERTa assigns each sentence exactly one economic agent and one sentiment label. This makes the output easy to aggregate, but it also imposes a simplified structure on Federal Reserve communication. Many sentences are simply not classifiable as talking about just one agent and may be economically relevant for several agents at the same time.

■ Households
 ■ Firms
 ■ Financial Sector
 ■ Government
 ■ Central Bank
▲ Positive sentiment
▼ Negative sentiment

Good afternoon, and welcome.▲ My colleagues at the Federal Reserve and I are dedicated to serving the American people.▲ We do this by steadfastly pursuing the goals that Congress has given us: maximum employment and stable prices.▲ We're committed to making the best decisions we can based on facts and objective analysis.▲ Today we decided to lower the interest rates for the third time this year.▲ We took this step to help keep the U.S. economy strong in the face of global developments and to provide some insurance against ongoing risks.▲ As I will explain shortly, the policy adjustments we have made since last year are providing—and will continue to provide—meaningful support to the economy.▲ We believe that monetary policy is in a good place.▲ The U.S. economy is in its 11th year of expansion, and the baseline outlook remains favorable.▲ The overall economy is growing at a moderate rate.▼ Household spending continues to be strong, supported by a healthy job market, rising incomes, and solid consumer confidence.▲ In contrast, business investment and exports remain weak, and manufacturing output has declined over the past year.▼ Sluggish growth abroad and trade developments have been weighing on those sectors.▼ Looking ahead, we continue to expect the economy to expand at a moderate rate, reflecting solid household spending and supportive financial conditions.▲ The job market remains strong.▲ The unemployment rate has been near half-century lows for a year and a half.▲ The pace of job gains has eased this year but has remained solid; we had expected some slowing after last year's strong pace.▼

Figure 3: Excerpt from the Federal Reserve press conference on 30 October 2019

This limitation is visible in the opening sentence, “Good afternoon, and welcome,” which is classified as positive firm sentiment. The sentence contains no substantive economic information, and the classification is therefore difficult to interpret economically. The underlying probabilities suggest that the model is uncertain: the sentence receives probabilities of 45 percent for firms, 27 percent for the central bank, 12 percent for households, 8 percent for the financial sector, and 8 percent for government. The assigned label is therefore the largest predicted class, but not a substantively meaningful economic category.

Another example is the sentence “Today we decided to lower the interest rates for the third time this year.” The sentence is classified as positive central-bank sentiment, which is understandable because the Chair is describing a policy decision by the Federal Reserve. Economically, however, the sentence is relevant for all five agents. For households, lower rates may be positive because they reduce borrowing costs and support consumption, but they may also reduce interest income for savers. For firms, lower rates can ease financing conditions and support investment. For the financial sector, the effect is ambiguous: lower rates may support asset prices and credit demand, but can compress net interest margins. For the government, lower rates reduce public-sector borrowing costs. For the central bank, the sentence may be positive if interpreted as an active stabilization measure, but it could also reflect concern about downside risks that made the rate cut necessary. The binary positive label therefore captures one dimension of the sentence, but not its full economic meaning.

This issue is closely related to the training procedure described by Pfeifer and Marohl (2023). In constructing the labelled dataset, they only labelled sentences where it was unambiguously clear that one and only one economic agent was being discussed. They explicitly note that some sentences, such as those concerning mortgages, may be relevant to households, firms, the financial sector, government, or the central bank depending on context. Sentences about more than one agent did not receive multiple labels in the training data. The resulting model is therefore well suited for producing standardized, scalable agent-specific measures, but it necessarily simplifies sentences whose economic implications cut across

several agents.

The excerpt also shows why this limitation does not invalidate the approach. Several classifications are economically intuitive. Sentences describing strong household spending, a healthy job market, rising incomes, and consumer confidence are classified positively, while sentences on weak business investment, exports, and manufacturing output are classified negatively. These examples suggest that the model can capture meaningful economic content when the sentence clearly refers to one agent or condition. The main limitation is therefore not that the model fails generally, but that its one-agent, binary-sentiment structure is less informative for generic, procedural, or multi-agent policy sentences.

Having assessed the usefulness and limitations of the sentence-level classifications, the next step is to examine whether the document-level sentiment measures contain useful economic information. For this purpose, I use the four-document rolling mean of agent-specific net sentiment and assess whether its movements are broadly interpretable in relation to major macroeconomic and policy episodes. Before turning to the time series, I first compare sentiment across minutes and press conferences.

Central Bank	0.708***	0.000	78	2012-09-13 to 2025-12-10
Government	-0.082	0.476	78	2012-09-13 to 2025-12-10
Financial Sector	0.413***	0.000	78	2012-09-13 to 2025-12-10
Firms	0.088	0.442	78	2012-09-13 to 2025-12-10
Households	0.682***	0.000	78	2012-09-13 to 2025-12-10
	Correlation	p-value	N	Sample period
* p < 0.10, ** p < 0.05, *** p < 0.01				

Figure 4: Correlation of rolling net sentiment between minutes and press conferences

The correlations are computed across matched FOMC meetings for which both minutes and press conference observations are available. For each agent, the table compares the four-document rolling mean of net sentiment across the two communication types.

The results show that sentiment in minutes and press conferences is strongly correlated for households and the central bank, with correlations of 0.682 and 0.708, respectively. Financial-sector sentiment also shows a positive and statistically significant correlation of 0.413. By contrast, firm sentiment is weakly correlated across communication types, and government sentiment is essentially unrelated.

Household and central-bank sentiment appear to capture common information across the two communication formats. This is plausible because both minutes and press conferences regularly discuss labor-market conditions, consumption, inflation, and the policy outlook. The positive correlation for financial-sector sentiment suggests that assessments of financial conditions are also reflected across both formats, although less strongly. The weak full-sample correlation for firm sentiment partly reflects the COVID-19 period. When observations from March to December 2020 are excluded, the firm-sentiment correlation rises from 0.088 to 0.204 and becomes weakly significant at the 10 percent level. This suggests that business-related sentiment is not entirely unrelated across minutes and press conferences, but

that the relationship is sensitive to crisis observations. A plausible interpretation is that during the COVID shock, the two communication formats served different roles. The government result should not be overinterpreted because government-related sentences are rare in both formats.

Overall, the correlations suggest that the document-level sentiment measures are not purely format-specific artifacts. For several economically central agents, especially households, the financial sector, and the central bank, sentiment moves similarly across minutes and press conferences. This supports the interpretation that the aggregated sentiment variables capture common underlying communication content, while still leaving room for format-specific differences.

Finally, the rolling sentiment series provides several examples of economically interpretable variation.

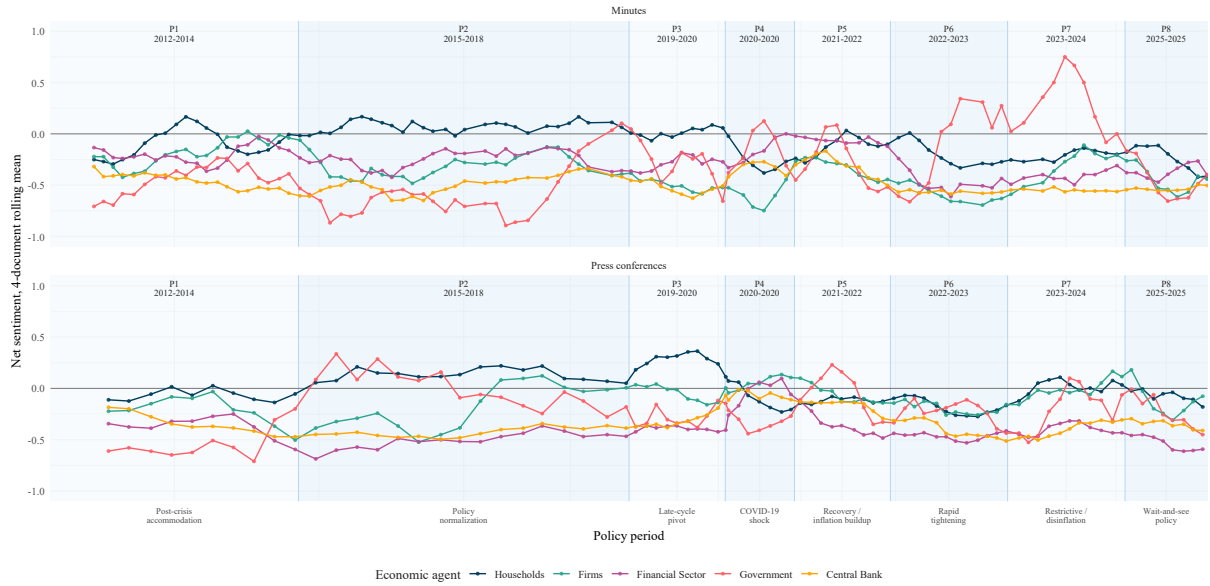


Figure 5: Rolling net sentiment

In the 2015–2018 normalization period, household sentiment in press conferences turns positive, consistent with the Federal Reserve’s communication about improving labor-market conditions and continued expansion. At the same time, central-bank sentiment remains negative in both formats, which is plausible in a period where communication focused on the cautious withdrawal of accommodation rather than on strongly positive policy news.

The COVID–19 shock is visible most clearly in the deterioration of household sentiment. In both minutes and press conferences, this category weakens around 2020, consistent with the sudden collapse in activity, employment losses, and stress in household and business conditions.

The inflation and tightening period is also reflected in the sentiment series. During 2021–2023, central-bank sentiment remains persistently negative in both minutes and press conferences, while financial-sector sentiment also weakens. This is consistent with a communication environment dominated by inflation risks, rapid rate increases, balance-sheet reduction, and tighter financial conditions.

Finally, the 2023–2025 disinflation and wait-and-see period shows a partial recovery in some sentiment categories, especially in press conferences. This fits a phase in which inflation had moderated and risks to the employment and inflation objectives had become more balanced, while the Federal Reserve remained cautious about declaring victory. The fact that sentiment improves only partially is important: the model does not simply classify late-sample communication as positive, but appears to capture the Fed’s

continued emphasis on uncertainty, data dependence, and remaining inflation risks.

Overall, these examples suggest that the rolling sentiment series contains more than mechanical variation. Broad movements in the LLM-based measures are consistent with major macro-policy episodes, especially normalization, the COVID-19 shock, the inflation surge, rapid tightening, and cautious disinflation. This supports the descriptive usefulness of the sentiment variables, while leaving the question of market relevance to the regression analysis.

A final point concerns the predominantly negative net sentiment in Figures 5 and 6. This pattern is consistent with Pfeifer and Marohl (2023)’s model documentation. Their labelled sentiment dataset contains more negative than positive observations, with 7,730 negative and 5,728 positive labels. They also note that communication related to specific economic agents may be more negative than central bank communication in aggregate. The negative net sentiment in this thesis should therefore not be interpreted as an anomaly. Rather, after each sentence is assigned one agent and one sentiment category, negative classifications are more frequent than positive classifications for most agent-document combinations. This likely reflects the fact that Federal Reserve communication often discusses risks, weaknesses, deteriorations, or policy-relevant constraints when referring to economic agents (Pfeifer & Marohl, 2023).

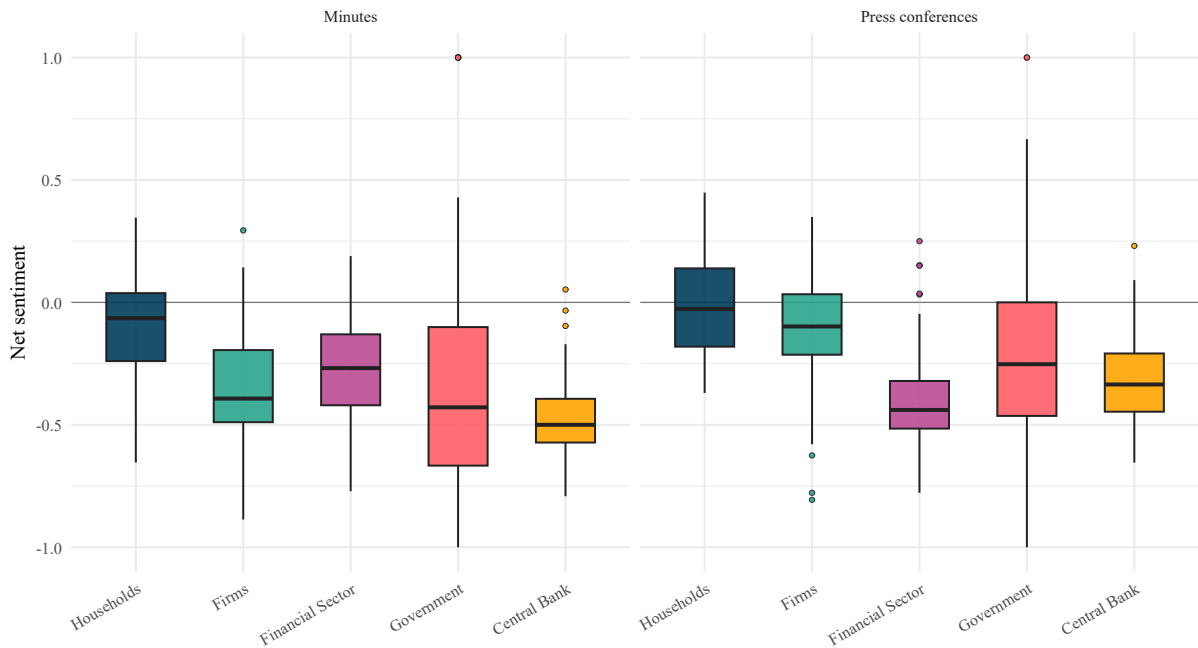


Figure 6: Agent net sentiment for minutes and press conferences

6 Empirical Results from Local Projections

The first hypothesis predicts that more positive household and firm sentiment should be associated with higher S&P 500 returns.

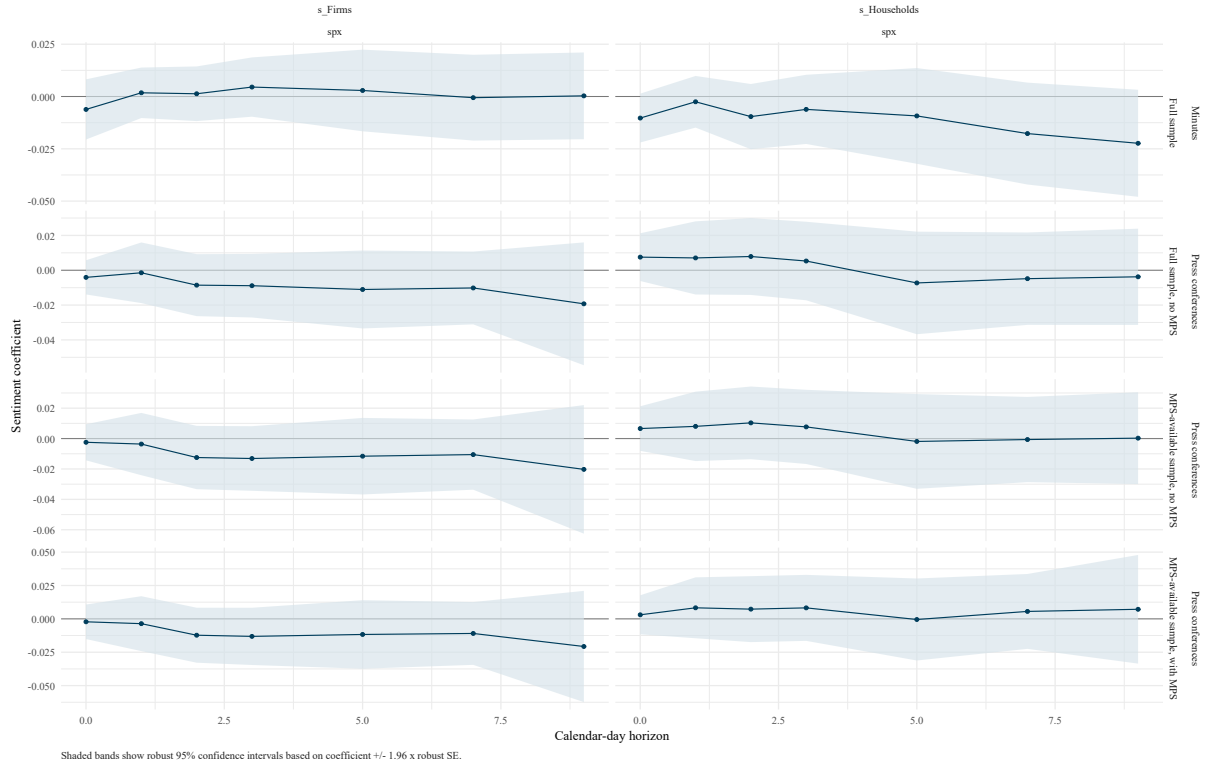


Figure 7: H1 coefficients

The results do not support this prediction. For FOMC minutes, firm sentiment is essentially uninformative for equity returns. The coefficient path remains close to zero across horizons and does not display a stable positive pattern. Household sentiment performs no better. In fact, the estimates are mostly negative, especially at longer horizons. The event-day coefficient is -0.010 and weakly statistically significant, while the nine-calendar-day coefficient is -0.022 and also weakly significant. These estimates go against the predicted sign, but they should not be overinterpreted because the significance is marginal and the pattern is not consistently strong across all adjacent horizons. The main conclusion is therefore not that household sentiment reliably lowers equity returns, but that the expected positive equity-market response is absent.

The press conference results reinforce this conclusion. Firm sentiment is mostly negative across horizons, especially in the longer return windows. Household sentiment is closer to zero and changes sign depending on the horizon and sample restriction. Adding the monetary-policy surprise control does not materially change the interpretation. In the MPS-available sample, the coefficient paths with and without the control remain similar, which suggests that the failure to find a positive real-economy sentiment effect is not simply due to omitted conventional policy surprises.

The failure of H1 is economically plausible. Positive real-economy sentiment in Federal Reserve communication does not represent pure good news for equities. It may signal stronger expected consumption, investment, and earnings, but it may also imply less need for monetary accommodation. In an equity-

pricing framework, these two channels can move in opposite directions: expected cash flows may rise, while expected discount rates may also rise. The resulting net effect on the S&P 500 can therefore be close to zero. The estimates are consistent with this ambiguity. They suggest that the equity market does not respond to household and firm sentiment in a simple “positive sentiment equals positive returns” manner.

The second hypothesis predicts a negative association between real-economy sentiment and two-year Treasury futures returns. This prediction follows from the inverse relation between Treasury futures prices and yields. If more positive household or firm sentiment makes investors expect a less accommodative future policy path, two-year yields should rise and two-year Treasury futures prices should fall.

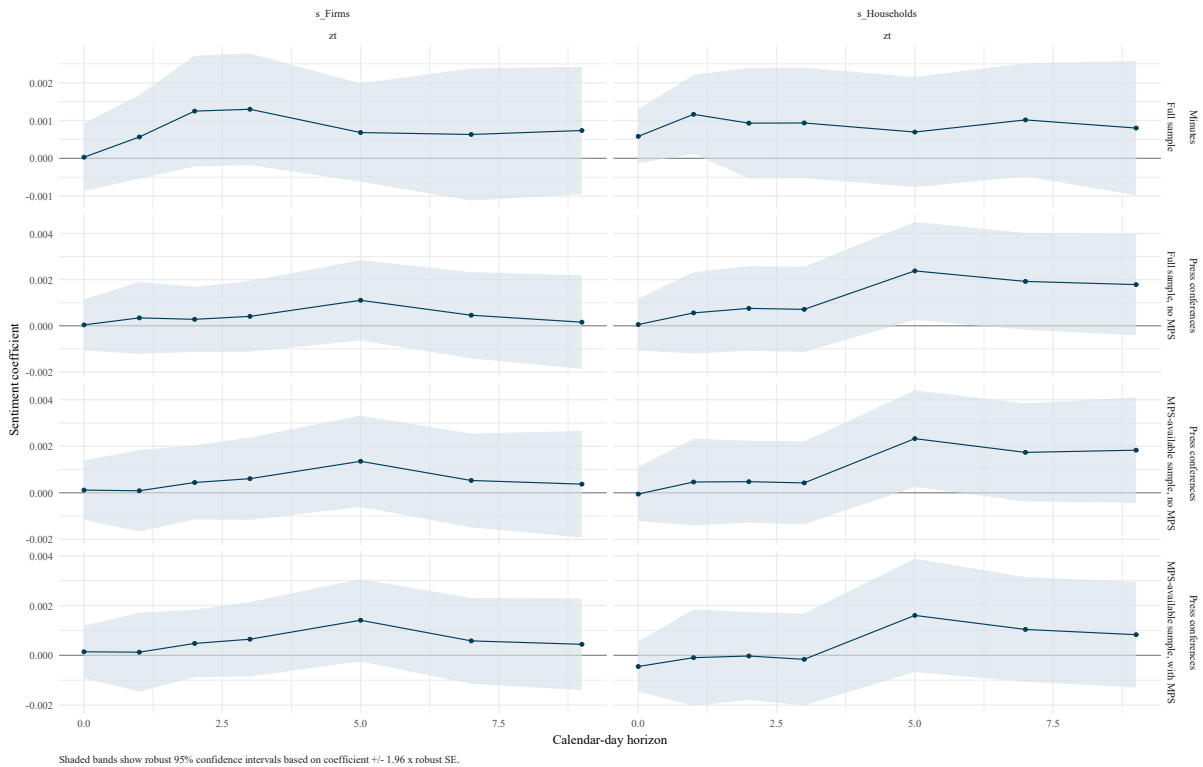


Figure 8: H2 coefficients

The estimated coefficients do not support this mechanism. Instead, the coefficients are mostly positive or close to zero.

For FOMC minutes, household sentiment is positively associated with two-year Treasury futures returns at most horizons. The one-calendar-day coefficient is 0.0012 and statistically significant at the five percent level. Firm sentiment also turns positive after the event day, with weakly significant coefficients at the two- and three-calendar-day horizons. These results are the opposite of the hypothesized sign. They do not suggest that more positive real-economy sentiment leads markets to price a tighter future policy path. If anything, the minutes results point weakly toward higher Treasury futures prices after more positive household or firm sentiment.

The press conference estimates show a similar pattern, especially for household sentiment. In the full press conference sample without monetary-policy surprise controls, household sentiment is positive across all horizons and becomes statistically significant at the five-calendar-day horizon, with a coefficient of 0.0024. It remains weakly significant at the seven-calendar-day horizon. In the MPS-available sample

without the control, the five-day coefficient remains positive and significant. Once the monetary-policy surprise control is added, the coefficient path still becomes positive at intermediate horizons, but the estimates are no longer statistically significant. Firm sentiment is also mostly positive, but its estimates are smaller and less precisely estimated.

These findings reject the narrow interest-rate interpretation behind H2. More positive household and firm sentiment is not associated with lower Treasury futures returns. A possible explanation is that real-economy sentiment in Fed communication does not only affect expected short rates. It may also affect perceived downside risk, policy uncertainty, or demand for safe and liquid assets. More reassuring communication about households or firms could support Treasury futures prices if markets interpret it as reducing uncertainty rather than as mechanically implying higher future rates. However, the evidence should still be described cautiously. The positive coefficients are not strong enough to establish a new alternative mechanism, but they are strong enough to reject the expected negative sign.

The third hypothesis produces the most interesting result.

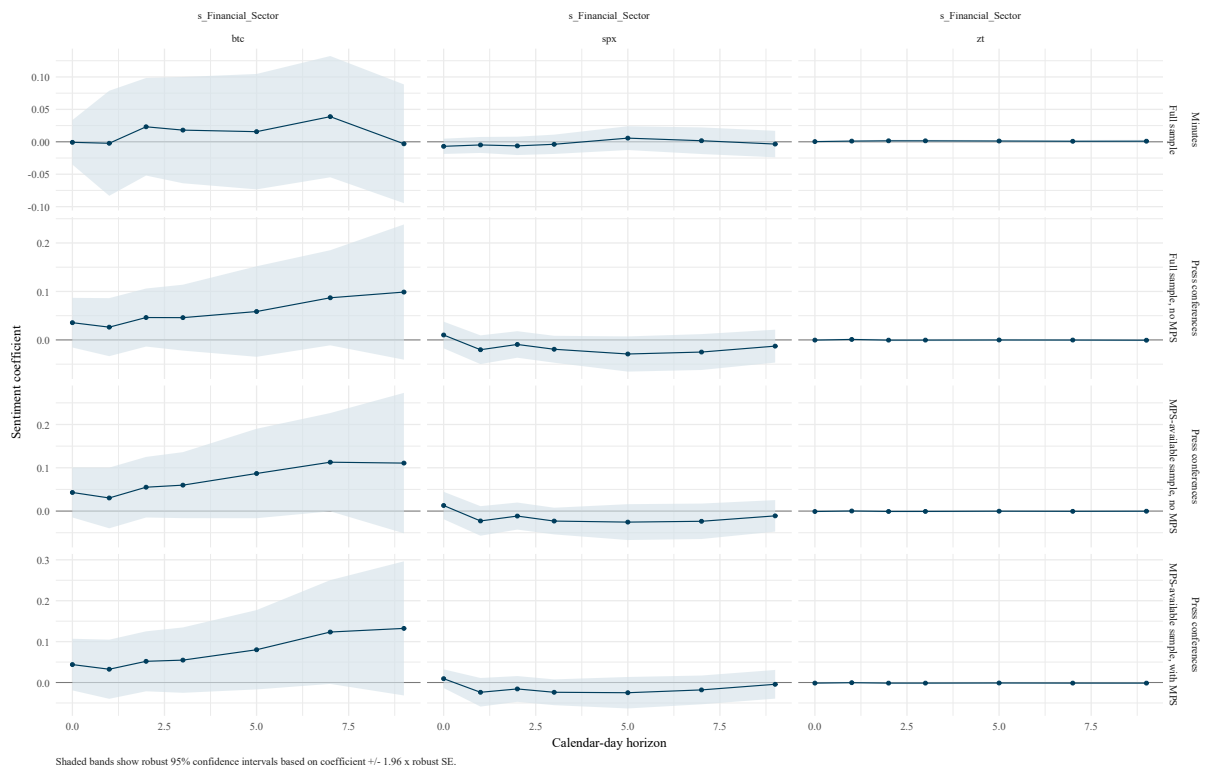


Figure 9: H3 coefficients

Financial-sector sentiment is not broadly priced across all risky assets, but it is visibly associated with Bitcoin returns after press conferences. This pattern is absent or much weaker for the S&P 500 and for two-year Treasury futures.

In the minutes sample, financial-sector sentiment does not show a strong relationship with Bitcoin or S&P 500 returns. Bitcoin coefficients are positive at several intermediate horizons, but the confidence intervals are wide. S&P 500 coefficients remain close to zero throughout. The only weakly significant minutes results appear for two-year Treasury futures at short horizons, where the coefficients are positive around the one- to three-calendar-day horizons. This means that the minutes results do not support the idea that financial-sector sentiment is especially relevant for risky assets.

The press conference results are different. Financial-sector sentiment is positively associated with Bitcoin returns across all press conference specifications. The coefficient path rises with the horizon and reaches its strongest point around the seven-calendar-day window. In the full press conference sample, the seven-day coefficient is approximately 0.087 and weakly statistically significant. In the MPS-available sample without the surprise control, the corresponding coefficient rises to approximately 0.113. With the monetary-policy surprise control included, the seven-day coefficient remains positive at approximately 0.123 and remains weakly significant. The persistence of the result after adding the MPS control is important because it suggests that the Bitcoin association is not merely a disguised response to the conventional policy-announcement surprise.

The same financial-sector sentiment measure does not produce a comparable S&P 500 response. In press conferences, the S&P 500 coefficients are mostly negative at intermediate horizons and statistically weak. This is an important result because it prevents an overly broad “risky asset” interpretation. If financial-sector sentiment simply captured general risk appetite, one would expect a clearer positive response in equities as well. Instead, the evidence suggests a narrower effect concentrated in Bitcoin. For two-year Treasury futures, the coefficients are very small and generally uninformative.

The Bitcoin result is economically plausible. As discussed in Chapter 2.3, Bitcoin has no cash flows and may be particularly sensitive to liquidity conditions, speculative demand, and shifts in macro-financial risk appetite. Press conferences are also the communication format in which the Chair can elaborate on financial conditions, market functioning, and risks in real time, making a positive financial-sector tone more immediately relevant for Bitcoin than the delayed, retrospective account in minutes.

At the same time, the result should be framed as suggestive rather than conclusive. The statistical significance is weak, and the confidence intervals are wide. The coefficient magnitude is also based on a one-unit change in net sentiment, which corresponds to a very large movement on a scale from -1 to +1. The economically relevant effect of a realistic sentiment change would therefore be smaller. Still, the pattern is coherent across the full press conference sample, the MPS-available sample, and the specification with the monetary-policy surprise control. Among all hypotheses, this is the clearest case where the sign, communication format, asset class, and horizon line up with a meaningful interpretation.

Taken together, the local projection results provide a disciplined but limited answer to the research question. The sentiment variables contain meaningful information, but this information is not broadly converted into predictable asset returns. H1 is not supported because household and firm sentiment do not predict higher S&P 500 returns. H2 is not supported because household and firm sentiment are not associated with lower two-year Treasury futures returns; the significant estimates point in the opposite direction. H3 receives partial support because financial-sector sentiment in press conferences is positively associated with Bitcoin returns, especially at longer horizons and after controlling for monetary-policy surprises.

7 Robustness Check

The motivation for this robustness check follows directly from the Bitcoin literature reviewed in Chapter 2.3. Earlier evidence suggests that Bitcoin was initially relatively detached from conventional monetary-policy transmission, while more recent work points toward stronger integration with broader financial markets. In particular, Bitcoin’s reaction to U.S. monetary policy appears to have changed over time, with the post-2020 period showing a clearer risky-asset response to contractionary monetary-policy shocks (Karau, 2023; Ma et al., 2022). The robustness check therefore examines whether the positive association between press-conference financial-sector sentiment and Bitcoin returns is also visible in the period in which Bitcoin may have become more exposed to macro-financial conditions.

I re-estimate the Bitcoin specification for Hypothesis 3 using observations from 2020 onward. For press conferences, I estimate the specification both without monetary-policy surprise controls for the 2020–2025 sample and, where the surprise data are available, with monetary-policy surprise controls for the 2020–2023 sample. For comparison, I also report the corresponding post-2020 estimates for FOMC minutes, although the main robustness check concerns press conferences.

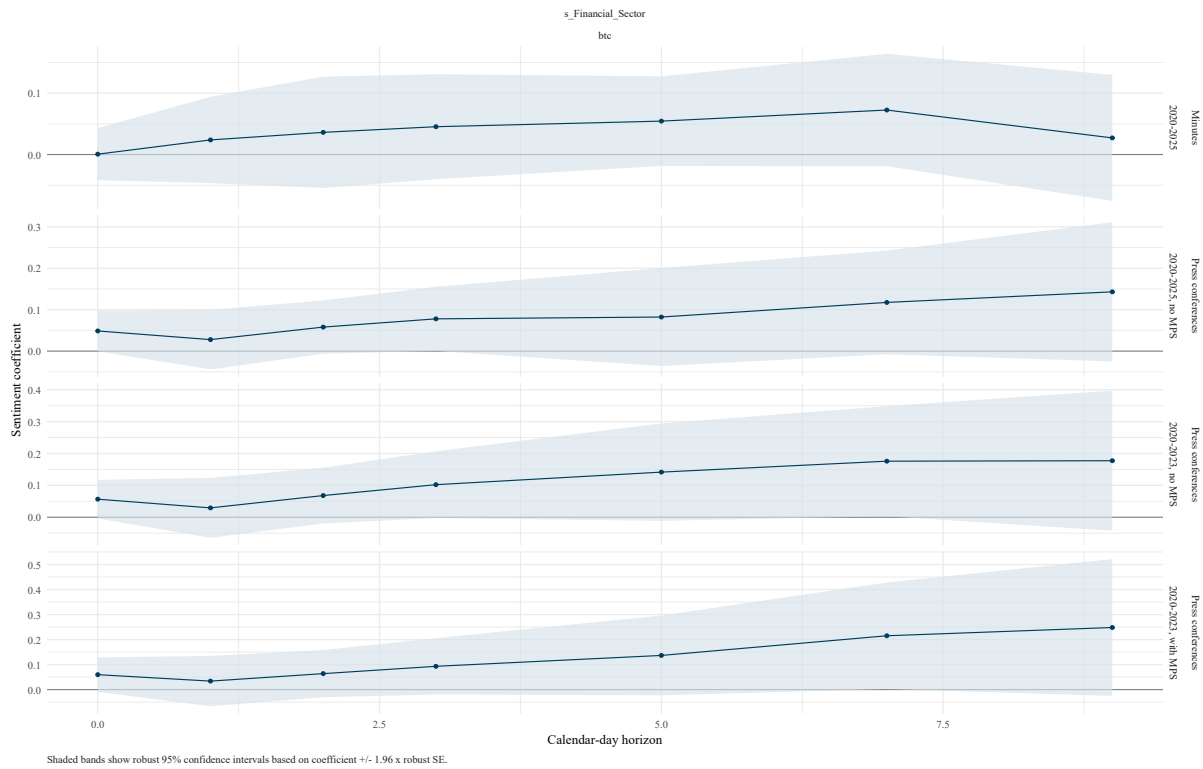


Figure 10: Robustness check: Financial-sector sentiment and Bitcoin returns after 2020

The results are broadly consistent with the main interpretation. For FOMC minutes, the coefficients are positive at most horizons and economically non-negligible. For example, the seven-calendar-day coefficient is approximately 0.073, which corresponds to a sizable cumulative Bitcoin return for a one-unit increase in financial-sector sentiment. However, the estimate is imprecise and statistically insignificant, so it does not provide strong evidence of a minutes-based Bitcoin response. This is in line with the main results, where minutes showed some positive Bitcoin coefficients at intermediate horizons, but the confidence intervals were too wide to support a clear relationship between financial-sector sentiment and

Bitcoin returns.

The press-conference estimates provide stronger support for the interpretation developed in the main results section. In the 2020–2025 sample without monetary-policy surprise controls, the coefficient on financial-sector sentiment is positive at every horizon. It rises from approximately 0.049 on the event day to 0.118 at the seven-calendar-day horizon and 0.143 at the nine-calendar-day horizon. The seven-day coefficient is weakly statistically significant, with a p-value of about 0.072. In the shorter 2020–2023 sample, where monetary-policy surprise controls can be included, the pattern becomes more pronounced. With the surprise control included, the coefficient reaches approximately 0.215 at the seven-day horizon and 0.248 at the nine-day horizon, with p-values of about 0.057 and 0.085, respectively. The positive coefficient path therefore remains visible after controlling for conventional monetary-policy surprises.

Overall, the robustness check strengthens the interpretation of Hypothesis 3, but only in a limited sense. The positive Bitcoin association is not purely a full-sample artifact; it is also visible in the post-2020 period and remains present when monetary-policy surprises are controlled for. The result is also concentrated in press conferences rather than minutes, which fits the interpretation that real-time explanatory communication by the Chair may be more relevant for Bitcoin than the more delayed and retrospective information contained in minutes. However, the evidence remains suggestive. The post-2020 sample is small, especially in the specification with monetary-policy surprise controls, and the confidence intervals remain wide. Moreover, the coefficient refers to a one-unit change in financial-sector net sentiment, which is large given that the sentiment measure ranges from -1 to +1. The robustness check therefore reinforces the main conclusion without changing its cautious interpretation.

8 Conclusion

This thesis examined whether agent-specific sentiment in Federal Reserve communication is associated with financial-market returns after FOMC minutes and press conferences. The empirical answer is deliberately modest. CentralBankRoBERTa produces sentiment measures that are economically interpretable at a descriptive level, but these measures do not translate into a broad and stable return response across asset classes. The main conclusion is therefore not that Federal Reserve sentiment generally moves markets. Rather, the results show that the market relevance of textual sentiment depends strongly on the category being measured, the communication format, the asset class, and the horizon over which returns are evaluated.

This distinction is important because the thesis stands at the intersection of two different tasks. The first task is textual measurement: whether a language model can extract structured information from central bank communication. The second task is economic interpretation: whether the extracted categories correspond to variables that can be linked to asset-pricing or monetary-policy transmission mechanisms. The descriptive results suggest that the first task is partly successful. The model produces plausible variation across communication formats and major macroeconomic episodes. Minutes and press conferences differ in ways that fit their institutional roles, and the sentiment series reflect episodes such as the COVID-19 shock, the inflation surge, the tightening cycle, and the later disinflation period. This supports the use of transformer-based models as tools for organizing complex policy texts.

The second task is more difficult. The empirical results show that agent-specific sentiment cannot be interpreted as a simple economic state variable. Positive household or firm sentiment is not consistently associated with higher S&P 500 returns, and it is not associated with lower two-year Treasury futures returns as predicted by a narrow interest-rate channel. This does not necessarily mean that Federal Reserve communication is irrelevant for these markets. A more plausible interpretation is that the sentiment categories are too broad relative to the mechanisms being tested. A positive sentence about households may refer to strong consumption, a resilient labor market, lower downside risk, or inflationary pressure. These implications can have different, and sometimes offsetting, effects on expected cash flows, discount rates, and policy expectations. The same issue applies to firm sentiment. Without a more explicit mapping from text categories to economic channels, the sign of the expected market response remains ambiguous.

This is the central limitation of the thesis. CentralBankRoBERTa classifies who is being discussed and whether the sentence is positive or negative, but it does not classify what the sentence implies for a theoretically defined variable. In that sense, the model’s categories are useful but not fully aligned with the empirical hypotheses. The label “financial-sector sentiment,” for example, may capture references to financial conditions, market functioning, banking-sector stress, credit availability, or risk appetite. These are related concepts, but they are not identical. Similarly, “positive sentiment” does not specify whether the sentence is positive for current conditions, future growth, inflation control, financial stability, or monetary accommodation. The economic interpretation of the model output therefore depends heavily on the conceptual structure that went into the original classification task.

Recent work points toward possible improvements. Kim et al. (2024) evaluate several language models on FOMC communication and show that model choice, fine-tuning, and task-specific design matter substantially for central bank sentiment analysis. Kim et al. (2025) go further by combining discourse

simplification with aspect-based sentiment analysis for FOMC minutes. This is closer to the type of measurement problem faced in this thesis because the relevant question is not only whether a sentence is positive or negative, but positive or negative with respect to a specific economic aspect. Audrino et al. (2024) provide another useful comparison. Their LLM-based uncertainty indices are built around economically interpretable categories such as monetary-policy uncertainty, financial-market uncertainty, and geopolitical risk. These approaches suggest that the next step in this literature is not merely to use larger language models, but to define the classification objective more tightly around economic theory.

Against this background, the empirical contribution of this thesis should be interpreted with clear boundaries. The results do not show that broad agent-specific sentiment categories are sufficient to identify the transmission of communication to asset prices. What they do show is more limited but still useful. First, agent-specific transformer-based sentiment can summarize Federal Reserve communication in a way that is descriptively meaningful. Second, most of these sentiment measures are not robustly priced in the return windows studied here. Third, the clearest market association appears in a narrower setting: financial-sector sentiment in press conferences is positively associated with Bitcoin returns, especially at longer horizons, a pattern that persists in the post-2020 subsample. Whether this reflects a genuine and stable communication channel or a sample-specific association remains an open question for future research.

The broader implication is that future research should treat textual classification as part of the economic research design rather than as a neutral preprocessing step. A stronger design would begin with a theory of the relevant transmission channel and construct labels that correspond directly to that channel—for example, classifying sentences by whether they imply stronger growth, tighter financial conditions, greater policy uncertainty, or lower downside risk.

Overall, this thesis shows both the promise and the limits of applying transformer-based sentiment models to central bank communication. The promise lies in the ability to extract structured and interpretable information from complex policy texts. The limit lies in the fact that economic interpretation requires more than accurate text classification. For financial-market applications, the categories produced by a language model must be tied to the mechanisms through which markets price information. Without that link, sentiment measures can be descriptively informative while remaining difficult to interpret as explanatory economic variables.

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A Auxiliary Aids Directory

Aid	Usage	Affected part/Example
ChatGPT	Creation and refinement of literature-search prompts	E.g. ("FOMC announcements" OR "Federal Reserve communication") AND ("high frequency identification" OR "event study") AND ("forward guidance" OR "policy statement" OR minutes OR speeches) AND ("interest rates" OR "bond yields" OR "financial markets")
ChatGPT, Claude	Discussion of research ideas, interpretation of results, limitations, and possible implications	Conceptual framing, interpretation of empirical findings, and discussion of limitations
Google Scholar, Google Scholar Labs, ChatGPT, Claude, HSG swisscovery, connected-papers.com, researchrabit.ai	Identification of relevant papers and journals; support in exploring and visualizing related literature	Literature review and bibliography preparation
GitHub Copilot, Codex, Claude Code	Assistance in drafting, debugging, and adapting web-scraping, data-processing, data-analysis, and visualization scripts	Code used for collecting Federal Reserve texts, preparing datasets, estimating regressions, and generating figures
EndNote	Citation management and bibliography organization	References and in-text citations
ChatGPT, Claude	Language editing and rephrasing to improve clarity, tone, conciseness, and readability	Selected passages throughout the thesis

Table 1: Auxiliary aids used in the thesis.

B Codebase

The code used for data collection, text processing, sentiment construction, return construction, and empirical analysis is available in the following GitHub repository:

`https://github.com/li2o/fomc-sentiment-returns.git`

The repository contains the scripts and supporting files used to reproduce the main computational steps of the thesis.

C Full Visualization of Press Conference Classification

This appendix reports the full sentence-level visualization of the Federal Reserve press conference on 30 October 2019, from which the excerpt in Figure 3 is taken. Each sentence is classified by Central-BankRoBERTa according to the economic agent it primarily refers to and the sentiment expressed about that agent. The visualization therefore provides the full context for the excerpt discussed in Chapter 5.

press_conference — 2019-10-30

Document type: press_conference | Meeting date: 2019-10-30 | Release date: | Document ID: speeches_00035
Source: <https://www.federalreserve.gov/mediacenter/files/FOMCpresconf20191030.pdf>

About this visualisation: Each sentence is classified by the *CentralBankRoBERTa* model along two dimensions: (1) the **economic agent** the sentence primarily refers to (colour-coded, see legend), and (2) the **sentiment** expressed about that agent (▲ **positive** / ▼ **negative**). Hover over any sentence for exact model probabilities. The document contains **279 sentences**.

■ Households ■ Firms ■ Financial Sector ■ Government ■ Central Bank ▲ Positive sentiment ▼ Negative sentiment

Good afternoon, and welcome. ▲ My colleagues at the Federal Reserve and I are dedicated to serving the American people. ▲ We do this by steadfastly pursuing the goals that Congress has given us: maximum employment and stable prices. ▲ We're committed to making the best decisions we can based on facts and objective analysis. ▲ Today we decided to lower the interest rates for the third time this year. ▲ We took this step to help keep the U.S. economy strong in the face of global developments and to provide some insurance against ongoing risks. ▲ As I will explain shortly, the policy adjustments we have made since last year are providing--and will continue to provide--meaningful support to the economy. ▲ We believe that monetary policy is in a good place. ▲ The U.S. economy is in its 11th year of expansion, and the baseline outlook remains favorable. ▲ The overall economy is growing at a moderate rate. ▼ Household spending continues to be strong, supported by a healthy job market, rising incomes, and solid consumer confidence. ▲ In contrast, business investment and exports remain weak, and manufacturing output has declined over the past year. ▼ Sluggish growth abroad and trade developments have been weighing on those sectors. ▼ Looking ahead, we continue to expect the economy to expand at a moderate rate, reflecting solid household spending and supportive financial conditions. ▲ The job market remains strong. ▲ The unemployment rate has been near half-century lows for a year and a half. ▲ The pace of job gains has eased this year but has remained solid; we had expected some slowing after last year's strong pace. ▼ Participation in the labor force by people in their prime working years has been increasing, and wages have been rising, particularly for lower-paying jobs. ▲ People who live and work in low- and middle-income communities tell us that many who have struggled to find work are now getting opportunities to add new and better chapters to their lives. ▲ This underscores for us the importance of sustaining the expansion so that the strong job market reaches more of those left behind. ▲ Inflation continues to run below our symmetric 2 percent objective. ▼ Over the 12 months through August, total PCE inflation was 1.4 percent and core inflation was 1.8 percent. ▼ Inflation pressures remain muted, and indicators of longer-term inflation expectations are at the lower end of their historic ranges. ▼ We're mindful that continued below-target inflation could lead to an unwelcome downward slide in long-term inflation expectations. ▼ However, against the backdrop of a strong economy and supportive monetary policy, we expect inflation will rise to 2 percent. ▲ Overall, we continue to see sustained expansion of economic activity, a strong labor market, and inflation near our symmetric 2 percent objective as most likely. ▲ While this has been our outlook for

quite some time, our views about the path of interest rates that will best achieve these outcomes have changed significantly over the past year. ▼ As I mentioned, weakness in global growth and trade developments have weighed on the economy and pose ongoing risks. ▼ These factors, in conjunction with muted inflation pressures, have led us to lower our assessment of the appropriate level of the federal funds rate over the past year. ▼ In both July and September, we reduced the target rate [range] for the federal funds rate by ¼ percentage point, and we did so again today, bringing the range to 1½ to 1¾ percent. ▲ The policy adjustments we've made to date will continue to provide significant support for the economy. ▲ Since monetary policy operates with a lag, the full effects of these adjustments on economic growth, the job market, and inflation will be realized over time. ▼ We see the current stance of monetary policy as likely to remain appropriate as long as incoming information about the economy remains broadly consistent with our outlook of moderate economic growth, a strong labor market, and inflation near our symmetric 2 percent objective. ▲ We believe monetary policy is in a good place to achieve these outcomes. ▲ Looking ahead, we'll be monitoring the effects of our policy actions, along with other information bearing on the outlook, as we assess the appropriate path of the target range for the fed funds rate. ▼ Of course, if developments emerge that cause a material reassessment of our outlook, we would respond accordingly. ▼ Policy is not on a preset course. ▼ Let me end with a few words about our technical monetary policy operations. ▼ In January, we made the key decision to continue to implement monetary policy in an ample-reserves regime. ▲ In that operating framework, we control the federal funds rate primarily by setting our administered rates, not through frequent interventions to actively manage the supply of reserves. ▼ In the transition to the efficient and effective level of reserves in this regime, we slowed the gradual decline in our balance sheet in March and we stopped it in July. ▲ In response to the funding pressures in money markets that emerged in mid-September, we concluded that it would be appropriate to maintain, over a time--over time, a level of reserve balances at or above the level that prevailed in early September of this year. ▼ To achieve this ample level, we announced on October 11 that we would purchase Treasury bills at least into the second quarter of next year as well as continue temporary open market operations at least through January. ▲ These actions are purely technical measures to support the effective implementation of monetary policy as we continue to learn about the appropriate level of reserves. ▼ They do not represent a change in the stance of monetary policy. ▼ In particular, our Treasury bill purchases should not be confused with the large-scale asset purchase programs that we deployed after the financial crisis. ▼ In those programs, we purchased longer-term securities to put downward pressure on longer-term interest rates and ease broader financial conditions. ▲ In contrast, increasing the supply of reserves by purchasing Treasury bills only alters the mix of short-term assets held by the public and should not materially affect demand and supply for longer-term securities or financial conditions more broadly. ▲ I will be happy to take your questions. ▲ So what we've said, to repeat, is that we see the current stance of policy as "likely to remain appropriate as long as incoming information about the economy remains broadly consistent with our outlook." And we say that, really, because both of the performance of the economy and the stance of policy. ▲ The performance of the economy has been--particularly the household sector--has been strong, has been resilient, with low unemployment, attractive levels of job creation, wages moving up, labor force participation moving up, household confidence, and solid gains in many measures of consumer spending. ▲ The manufacturing sector, particularly manufacturing, and also investment and exports have been weaker. ▼ But, overall, we've seen

moderate growth, a strong labor market, inflation moving up. ▲ We see the outlook as for more of the same. ▼ We also see the risks to the outlook as perhaps having moved in a positive direction over the course of this intermeeting period, although, some remain--some things remain to be seen. ▼ So that's the economy. ▲ Turning to policy, we've moved the stance of policy over the course of the year to a more accommodative stance, and, after the cuts at the last three meetings, the federal funds rate is now in a range between 1½ and 1¾ percent. ▲ We feel that policy is well positioned to support the outlook that I described. ▲ So you asked what it would take, you know, to move, and, as I mentioned, we're going to be watching all factors, And if developments emerge that cause a material reassessment of our outlook, we would respond accordingly. ▼ But that's what it would take: a material reassessment of our outlook. ▼ We'll be looking at a full range of data about the economy and about the risks to the outlook, and I've given you a sense of what our outlook is. ▼ It's for moderate growth, a strong labor market, and inflation near our 2 percent objective. ▲ If something happens to cause us to materially reassess that outlook, that's what would cause us to change our views on the appropriate stance of policy. ▼ So, in terms of risks, what I was referring to there--the principal risks that we've been monitoring have been, really, slowing global growth and trade policy developments as well as muted inflation pressure. ▼ So I was really referring there to trade developments. ▼ We have that phase-one potential agreement with China, which, if signed and put into effect, could have the effect of reducing trade tensions and reducing uncertainty, and that would bode well, we think, for business confidence and perhaps activity over time. ▲ So that has the potential for being an improvement in the risk picture. ▲ Brexit, I would say, as well--it appears that the risk of a no-deal Brexit seems to have materially declined. ▲ I think on both situations there's plenty of risk left, but I'd have to say that the risks seem to have subsided. ▼ You asked about business investment, and that's right. ▲ Business investment has been weak, and today's reading was weak as well. ▼ And it was broad across equipment and other parts of--all parts of business fixed investment were weak. ▼ That's consistent with what we've seen, but that's the economy we've had this year. ▲ What we've had is an economy where the consumer is really driving growth, and, you know, personal consumption expenditures were almost 3 percent in this quarter--in this first reading for the quarter. ▲ So, overall, we see the economy as having been resilient to the--you know, the winds that have been blowing this year. ▲ Yes, so that is a risk. ▼ That's a risk that we've been monitoring, but we don't see it yet. ▼ We don't see rising initial claims or layoffs or anything like that. ▼ But the risk you mention of the weakness in the manufacturing--in export investment, business investment, parts of the economy getting into the consumer side of the economy, we really don't see that. ▼ What we continue to see is good job creation. ▲ Unemployment has declined again, and the household survey is now at a 50-year low--has been, you know, very close to 50-year lows for 18 months now. ▲ So it's very positive. ▲ The consumer-facing companies that we talk to in our vast network of contacts report, you know, that consumers are doing well and are focused on, you know, the good job market and rising incomes, and that's their principle focus. ▲ So that is the thing that's pushing the economy forward, and it doesn't seem to have been affected so far by weakness in the other areas. ▲ So the reason why we raised interest rates is because--generally is because we see inflation as moving up--or, in danger of moving up significantly, and we really don't see that now. ▼ Inflation moved down in the first quarter of this year. ▲ We thought that that was due, to some extent, to idiosyncratic or transient factors. ▼ That turns out to have been the case, so it's moved back up. ▲ But it seems to be settling in below 2 percent. ▼ So we really don't see that risk. ▼ And

inflation expectations have also kind of moved down and sideways--both surveys and market based over the course of this--of, really, the recent months. ▼ And, you know, we think that inflation expectations are very important in driving actual inflation, and we're strongly committed to achieving our 2 percent inflation objective on a symmetric basis. ▲ We think it's essential that we do that. ▲ So we're not thinking about raising rates right now. ▼ There obviously will be times in the future where that will be appropriate. ▲ What we're really thinking now is that our current stance of policy is appropriate and will remain so as long as the outlook is broadly in keeping with our expectations. ▲ So we think that the current stance of policy is likely to remain appropriate--likely to remain appropriate--as long as incoming information about the economy is broadly consistent with our outlook, which is a positive one of moderate economic growth, strong labor market, and inflation moving close to 2 percent. ▲ That's--so that's what we're saying about that. ▼ We also say, of course, if developments emerge that cause a material reassessment of that outlook, we would respond accordingly. ▼ So that's really how we're thinking about it. ▼ I can't point to one data point or one thing that would change our minds. ▼ It really--it would be a reassessment of--a material reassessment of our overall outlook, which is what I described. ▼ Well, there's--there's a lot of, you know, practice and science and history, in terms of how much you put in the actual postmeeting statement as opposed to what we say in the press conference statement. ▼ It's a--you know, it's a judgment call. ▼ You know, I'm saying it now, so-- CHAIR POWELL. ▼ So, as I mentioned, we do think that inflation expectations are-- they're quite essential, quite central in our framework of how we think about inflation. ▼ We need them to be anchored in a level--at a level that's consistent with our symmetric 2 percent inflation goal. ▼ And we think that we need to conduct policy in a way that supports that outcome. ▲ That's what we're doing now. ▲ We're also, as part of our review, looking at potential innovations, changes to the way we think about things, changes to the framework that would lead us--that would be more supportive of achieving inflation on a 2 percent--on a symmetric 2 percent basis over time. ▲ That's at the very heart of what we're doing in the review. ▼ It's too early to be announcing decisions. ▼ We haven't made them yet. ▲ But we're in the middle of thinking about ways that we can make that symmetric 2 percent inflation objective more credible by achieving symmetric 2 percent inflation. ▲ And it comes down to using your policy tools to achieve 2 percent inflation, and that is the--that is the thing that must happen for credibility in this area. ▼ So we're committed to doing that. ▲ So we're in the middle--we're really quite in the middle of it now, and my thinking is still that it will run into the middle of next year. ▼ These are--you know, these changes to monetary policy frameworks happen--they don't happen really quickly, let's say. ▼ Inflation targeting took many years to evolve. ▼ I don't think we'll take many years here. ▼ I think we'll wrap it up around the middle of next year, would be my guess. ▲ I have some confidence in that. ▲ Rate hikes for next year. ▼ So I would just say that a--if we were to have a sustained reduction in trade tensions--broad reduction in trade tensions and a resolution of these uncertainties, that would bode well for business sentiment, which is--trade uncertainty's been weighing on business sentiment, in our judgment, in the judgment of many analysts. ▲ And, ultimately, it could affect activity. ▼ I wouldn't expect that the effects on activity or confidence would be immediate, or--would there be immediate effects in economic activity. ▼ I think it would take some time after the recent things, but I do think it would be quite positive over time. ▲ You come back to the question of raising rates. ▼ So that's really about inflation. ▼ And you know, we haven't yet--we just touched 2 percent core inflation, to pick one measure--just touched it for a few months, and then we've fallen

back. ▼ So I think we would need to see a really significant move up in inflation that's persistent before we would consider raising rates to address inflation concerns. ▼ Well, I think we're doing that. ▲ I mean, I think we're keeping--I think we've made very substantial adjustments to policy over the course of this year, if you think about it. ▼ We entered the year expecting some further rate increases, we went to "patient," now we've done three rate cuts. ▼ It's a very substantial shift, and the effects of it will be felt over time. ▼ So we feel like those shifts are appropriate to support exactly the outcomes you're talking about, which are a continuing strong labor market, continued strong job creation. ▲ It was--so we've said it was about three things. ▼ It's been about the slowing global economy. ▼ You have a synchronized slowdown in economic activity around the globe. ▼ It's been building for just about 18 months now, and that's having an effect on CHAIR POWELL. ▼ So, on the "how long," what we've said is that we expect bill purchases to continue at least into the second quarter of next year. ▲ And we've said that temporary open market operations we expect to continue, I think, at least until the end of January, I believe. ▲ Yes, through January. ▲ In terms of the causes, so there's a lot of forensic work going on by us and by market participants and all kinds of analysts. ▼ And, you know, one thing is that we think we need reserves to be back up to the level--the minimum level of reserves that we can have during the various fluctuations that you see with reserves is something like \$1.45 trillion or a little higher, and that's the level in early September. ▼ So we're going to be adding reserves to get back to that place. ▲ There are also--it may be--one thing that was surprising about the episode was that liquidity didn't seem to flow as one might have expected. ▼ We had surveyed the banks carefully about what was their lowest comfortable level of reserves, and many banks that were well above that level did not take that excess cash and invest it in the repo market at much higher rates. ▼ They just--they didn't do that, and so the question is why. ▼ And are there things that we can do that would--adjustments that we could make that would allow liquidity to flow more easily in the system without in any way sacrificing safety and soundness or financial stability? ▲ So we're looking at those. ▲ Those are not things that can happen that can really address the situation in the short term, but those are a range of things that we're looking at as well. ▼ Honestly, it's really the--there isn't any one factor. ▼ It's really the whole range of economic data and also what we see as--about our stance of policy, which we believe is the appropriate stance. ▲ It's really both of those things. ▲ Again, if you look at the performance of the economy through this, the consumer sector has been quite resilient. ▲ Witness today's GDP report. ▲ And it's just--again, all the things I went through earlier suggest that that sector continues to be strong. ▲ We know that the manufacturing, investment, trade, export sector has been weak. ▼ That continues, but nonetheless, overall, we've got an economy that is showing moderate growth, and we think that today's setting--we're now at 1.5 to 1.75 percent. ▼ That is, we believe, an accommodative level and a level that will support that outlook that we have. ▲ You know, we don't see any evidence of that. ▲ What we see is--what we see is that the consumer sector continues strong. ▲ Again, low initial claims. ▼ It's all--the data we monitor, obviously, is a very broad range of data. ▲ We also see now, I think, more clearly the effects of more accommodative monetary policy on various kinds of consumer activity. ▼ You're seeing strong durable goods sales. ▲ You're seeing housing now contributing to growth for the first time in a while--two-tenths, I think, this quarter. ▲ And you're seeing retail sales--the retail sales number, the PCE number, as I mentioned, in today's release was 2.9 percent.⁴¹ ▲ And I think, more broadly, monetary policy is also supporting household spending and home buying by keeping the labor market strong, keeping workers' incomes rising,

and keeping consumer confidence at high levels, where it currently is. ▲ Well, I think monetary policy works through the channels that we understand. ▲ I think the effects are clear in what we think of as the interest-rate-sensitive sectors, which are the ones--some of the ones that I mentioned. ▼ I think it's--I think interest rates are a factor in business investment, but I don't believe they're the main factor and main driver. ▲ And I think, you know, what one would like to see for--to support greater manufacturing activity, business investment, and exports would be a global recovery. ▲ And, you know, there's a lot of monetary policy accommodation and some fiscal accommodation, maybe, in the global economy now. ▼ Not here, but perhaps elsewhere. ▼ And that should turn--that should support growth in the global economy. ▲ And, again, resolution of our trade issues would contribute to that over time as well. ▲ We think--we think that Fed Listens has been a great success for us, and I'm sure we will repeat it. ▲ In fact, I would imagine that this entire monetary policy review will become institutionalized and be done every few years. ▼ I would say this: In the Beige Book, we talk to educational institutions, health-care institutions, community groups, labor groups--it's not just businesses. ▲ And, by the way, all of those groups are also represented on the boards of our regional banks. ▲ We also meet quite regularly at the Board with representatives of low- and moderate-income communities. ▲ And so we're--we're very conscious of--you know, that we represent all Americans, and we need to hear all of their perspectives. ▲ That's why we--you know, we talk about how low the unemployment rate is, the aggregate unemployment, but we also talk about groups that haven't experienced that yet. ▲ And so we're--we try to remind ourselves that we serve everybody. ▲ So that is--our outlook overall is for moderate growth of around 2 percent, which is pretty close to trend. ▲ That's our outlook: could be better, could be worse. ▼ You know, it's--you never say you've achieved it. ▲ But that's our outlook. ▼ And we feel like our current stance of policy is appropriate as long as that remains broadly our outlook. ▲ I think this seems to be the right movement. ▲ You can see that we had our usual range of perspectives, range of views. ▲ But, in the end, we had a strong vote in favor of this action. ▲ I strongly believe it's the right action. ▲ I strongly believe that the actions we've taken over the course of this year have been the right things to do for this economy and are supporting economic growth and will do so in the future. ▲ You know, it's--monetary policy is famously a blunt instrument. ▼ So we can't--obviously, we can't raise interest rates some place and lower them other places. ▼ But we're very conscious. ▼ The first thing to know is to understand that, that--and we've called that out, as you, I'm sure, know, in one of our recent Monetary Policy Reports to Congress. ▼ The sort of rural-urban disparities in--you know, in employment, in growth, in all kinds of things, in health outcomes. ▼ You know, it's a big and growing difference--set of differences. ▼ We don't really have, you know, the tools to address that, but we call things out that are important to our economy that we may not be--that we might have--may not have the right tools to best address them, but we call them out because, you know, we're spending all of our time with the economy, and those are issues that maybe have to be addressed by legislatures and, you know, by--at the national level and at the state level. ▼ I'm not asking Congress to do anything right now. ▼ I sometimes have noted that it would be appropriate. ▲ Let me put it this way: We do what we do, which is, we're assigned a job, which is maximum employment and stable prices. ▲ We do financial stability. ▲ We do bank regulation. ▼ The U.S. economy faces significant longer-term challenges around potential growth, around labor force participation, around disparities of income and wealth, around all kinds of things. ▼ And those are issues for Congress, and they're not issues for us. ▼ But if you really want the U.S. economy to

be all it can be and raise the potential growth rate of the United States, you need--you need proper monetary policy. ▲

But, really, it isn't monetary policy. ▼ Really, it's fiscal policy that supports, you know, inclusive growth. ▲ Yes, I know, that's--I wouldn't--that's not how I would characterize it. ▼ I would just--I noted that it may be that there's been progress there toward-- away from bad outcomes and confrontational outcomes, and it seems like there's the makings of a possible settlement there. ▲ I didn't say we're baking that in or anything, but I'm just noting that that's the case. ▲ As with Brexit--it's still quite uncertain what will be the result of the election and what the ultimate result will be. ▼ But I would say that the tail risk of a nonnegotiated no-deal Brexit seems to have decreased, just as I would say that the situation in our trade negotiations with China seems to have taken a step closer to resolution. ▲ That's all I'm saying. ▼ That's one factor among many, many factors that factor into our assessment of the outlook. ▼ And if we see--if we have--if we--if things happen that cause us to materially reassess the outlook, change our assessment of the outlook in a material way, then we would act accordingly. ▼ So, not for us to say what the appropriate labor arrangements will be. ▼ I will just say, on the General Motors strike, it's--it's likely to have taken away a couple of tenths of growth this quarter. ▼ That's likely to come back, maybe, over the first half of next year. ▲ And I think it's good to see it settled. ▲ But I'm not going to comment on what the appropriate labor arrangements are. ▼ That's a little bit far afield from our task here. ▼ Sorry, say again what your first question was? ▼ I wrote it down, I can't-- CHAIR POWELL. ▼ So if you look at where the federal funds rate is trading, it should be in the middle of, or maybe the lower half of, the range of 150 to 175. ▼ So that means the real rate is probably modestly below zero. ▼ I think my own sense would be that that's somewhat accommodative policy. ▲ I would say, though, that there's a range of plausible estimates of what the neutral rate of interest is, and I think many of those who make such estimates have moved their estimate down over the course of many years, and that process continues. ▼ But, nonetheless, that seems to me to be very likely to be an accommodative stance of policy, and an appropriate stance, given the situation we're in with continuing downside risks and--that I mentioned. ▼ So we monitor financial stability risks very carefully all of the time. ▼ It's what we do since the financial crisis, as I've mentioned before. ▼ Currently, we don't see large imbalances. ▼ This long expansion is notable for the lack of large financial imbalances like the ones we've seen certainly before the crisis happened. ▼ So we have a four-part framework, I'll quickly mention. ▼ The first is leverage in the financial system, which is low by historical standards. ▼ The second is funding risk, which is the risk of runnable funding, and that risk is also quite low for banks but also for, you know, the nonbanking financial sector. ▼ If you look at asset prices, we see some high asset prices but not broadly across a range. ▼ We don't see bubbles and that kind of thing. ▲ And that leaves the fourth, which is leverage in the nonfinancial sector, and that's households and businesses. ▼ So with households, again, we don't see leverage. ▼ We see them actually getting in very good shape financially in the aggregate. ▲ Obviously, plenty of households are not in great shape financially, but in the aggregate, the household sector is in a very good place. ▼ That leaves businesses, which is where the issue has been. ▼ Leverage among corporations and other forms of business, private businesses, is historically high. ▲ We've been monitoring it carefully and taking appropriate steps. ▲ So that's what I would say, but it's--corporate debt is one part of a larger part of our framework, and it is something that we're paying quite a bit of attention to. ▼ And it's been part of the last couple of Shared National Credit exams, and we've been monitoring it carefully and taking appropriate action. ▼ So we're--I think, again, I think the most important basic

thing is to get the level of reserves back up so that--reserves move up and down with some volatility. ▼ We don't want them to move below the level they were at in the beginning of September, which is, again, between \$1.45 and \$1.5 trillion. ▼ That's the main thing, that's the first thing. ▼ And we're on a path to do that, between our temporary open market operations and also our bill purchases. ▲ In addition to that--in addition to that, we're looking at--there are--it's a big, complicated marketplace, and there was the--one of the surprises, as I mentioned, was that banks that had told us that their lowest comfortable level of reserves was here. ▼ They were well above that, and yet they didn't deploy that liquidity when there seemed to be great opportunities to do that. ▼ And so, we're doing careful analysis of that. ▼ You ask, would we lower capital and liquidity requirements because of this? ▼ I don't think that's where this goes. ▼ I think where it may go, though, is to look at, for example, intraday liquidity, which used to be a common thing. ▼ It used to be a common thing for banks to have intraday liquidity from the Fed, what are called "daylight overdrafts." That's something we can look at. ▼ Also, there are just a few technical things that we can look at that would perhaps make the liquidity that we have--which we think is ample in the financial system--move more freely and be more liquid, if you will. ▲ And those are things we would do, but only if we can do it without compromising safety or soundness or financial stability. ▼ So we--we, of course, have watched the situation in Japan and now the situation in Europe, and we--we note that there are significant disinflationary pressures around the world. ▼ And we don't think we're exempt from those. ▼ But, of course, if you look at our current inflation performance, it hasn't been anything like what we've seen in those other places. ▼ But we don't think--again, we don't think we're exempt from those pressures, and we are, therefore, strongly committed to having inflation expectations anchored at the level that is consistent with the symmetric 2 percent inflation objective. ▼ That's what we're committed to, and we'll use our tools to achieve. ▲ So we take--we take the risk very seriously. ▼ It's not--the risk isn't that inflation might run a couple tenths below 2 percent. ▼ The risk is that what we've seen is other economies getting on a disinflationary path, that it's been very hard for them to get off. ▼ Once inflation expectations start sliding down, inflation moves down. ▼ You have less--interest rates move down as well, because there's an inflation component in interest rates. ▼ And, you know, we think that the right thing to do is to do what we can now to hold and really move inflation expectations up so that they're squarely and firmly anchored at a level that's consistent with 2 percent inflation. ▲ You know, it's a--it's an issue, but it's not an issue that we're currently working on and--or reaching a decision on or anything like that. ▼ The question of the longer-run maturity composition of the balance sheet is a--is a big one, and it's one we'll return to over time, but not--not imminently. ▼ You know, I'm going to maintain my longtime practice of not--not commenting on anything any elected official would say, but thank you. ▼

Document-level summary

Agent	Sentence count	Share	Avg. confidence
■ Households	44	15.8%	0.791
■ Firms	51	18.3%	0.720
■ Financial Sector	67	24.0%	0.764
■ Government	14	5.0%	0.644

■ Central Bank	103	36.9%	0.743
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